From Recognition to Reasoning: Edge-Frequency Chain-of-Thought for Remote Sensing Object Detection

Xun Li, Yuzhen Zhao, Baoxi Yuan, Yang Zhao, and Ruijuan Yao

***Abstract*—The main bottleneck of current object detectors does not lie in weak feature representation. It lies in the lack of structured reasoning. Most detectors follow a recognition-style pipeline. They map features directly to detection results. This design ignores structural relations inside visual information. As a result, model stability and robustness degrade in complex scenes. This article proposes edge-frequency chain-of-thought (EF-CoT), a structured reasoning paradigm for object detection. EF-CoT reformulates detection as a stepwise reasoning process. It jointly models domain-specific tokens, spatial priors, and progressive reasoning. The model analyzes visual evidence before making predictions. Based on EF-CoT, this article designs the triple-branch guided frequency alignment and edge-aware network (TGFAENet). TGFAENet builds edge-frequency-aware structured tokens. Token-level reasoning then models the interaction among global structure, local details, and boundary cues. The Gaussian attention spatial pyramid pooling fusion (GASPPF) module is further introduced. GASPPF uses dual-branch spatial pyramid pooling and Gaussian attention (GA) to model continuous spatial priors. This module guides the detector toward potential target regions. By combining structured token modeling, spatial awareness, and chain-style reasoning, the proposed method builds a unified perception–analysis–decision detection framework. Experiments on complex remote sensing scenes show clear performance gains. The gains are more visible on small-object detection. These results support the shift from direct recognition to diagnostic reasoning in object detection.**

***Index Terms*—Chain-of-thought (CoT) reasoning, feature fusion, multistage representation learning, object detection, remote sensing imagery.**

I. Introduction

In RECENT years, deep convolutional networks and Transformer architectures have advanced object detection methods [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12]. However, complex remote sensing scenes still expose several limitations. Small targets occupy only a few pixels. Background clutter often dominates the image. Object boundaries are frequently blurred [13], [14], [15]. These factors make detectors confuse targets with background regions. They also make high-frequency textures resemble real object structures. As a result, false positives, missed detections, and localization errors still occur.

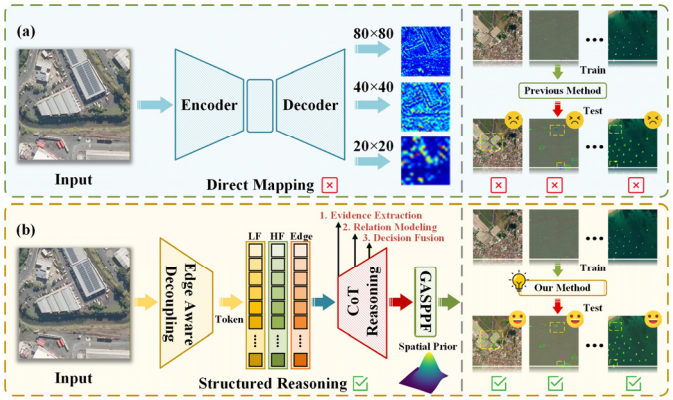

Fig. 1. Comparison between conventional detection paradigms and the proposed EF-CoT workflow. Traditional methods perform direct recognition without explicitly modeling intermediate reasoning steps. In contrast, EF-CoT follows a human-like perception process, progressively capturing global structure, refining local details, and integrating boundary cues, thus enabling an explicit CoT reasoning mechanism of "analyze-reason-decide." (a) Conventional paradigm. (b) ET-CoT paradigm (ours).

Most existing detectors follow a discriminative paradigm. They predict object categories and locations directly from feature representations. This end-to-end mapping is effective. Yet, it mainly relies on implicit feature learning. Global structure, local detail, and boundary cues are often encoded together in a shared feature space. Intermediate visual evidence is not explicitly modeled. Human visual perception follows a different process. It usually proceeds from global structure perception to local detail analysis and then to boundary cue integration. This process can be viewed as a chain-of-thought (CoT) mechanism [16], [17]. As shown in Fig. 1, existing detection methods mostly remain at the direct recognition stage. They rarely describe this step-by-step analysis and reasoning process explicitly [18], [19].

Existing studies have explored CoT and step-by-step reasoning in language understanding, multimodal understanding, and visual reasoning. Representative examples include visual

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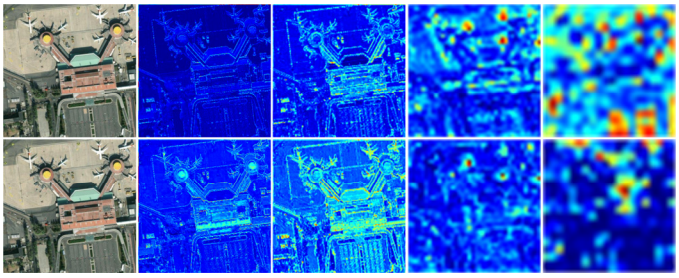

Fig. 2. Multiscale feature response comparison between the (top) baseline and (bottom) proposed EF-CoT paradigm. The baseline exhibits diffused activations scattered over background regions, failing to highlight small objects. In contrast, EF-CoT produces compact and structured responses, enhancing targets while suppressing noise. This shift indicates a transition from direct recognition to explicit CoT reasoning via progressive evidence refinement in feature space.

CoT, planning-based reasoning with language models, relation reasoning, and graph-structured reasoning [20], [21]. However, most of these methods focus on general visual understanding, visual question answering, or high-level object relation modeling. They do not match the core challenges in remote sensing object detection. Small targets occupy limited image regions. Complex backgrounds introduce dense interference. Boundary cues are often weak. Low-frequency structure and high-frequency detail are also strongly coupled.

To address this gap, this article proposes an edge-frequency CoT (EF-CoT) paradigm. EF-CoT reformulates remote sensing object detection as a process that combines structured token modeling with step-by-step reasoning. The CoT in this article does not refer to text reasoning in language models. It refers to explicit intermediate states in the visual feature space. Detection is then performed through three ordered stages: evidence extraction, relation modeling, and decision fusion. As shown in Fig. 2, baseline features show diffuse responses and strong background interference. EF-CoT produces more compact and structured target activations. This result shows that EF-CoT analyzes and integrates key visual evidence before prediction.

EF-CoT contains two core components. The first component is a domain-specific tokenizer. It decomposes input features into low-frequency tokens, high-frequency tokens, and edge tokens. These tokens describe global structure, local detail, and geometric boundaries, respectively. The second component is a token-level chain reasoning mechanism. It integrates structured visual information through evidence extraction, relation modeling, and decision fusion. Based on this paradigm, this article further builds the triple-branch guided frequency alignment and edge-aware network (TGFAENet). TGFAENet enhances the representation of small targets and weak-boundary targets through edge-frequency decomposition and cross-branch interaction. This article also designs the Gaussian attention spatial pyramid pooling fusion (GASPPF) module. GASPPF introduces continuous spatial priors through multiscale spatial pyramid pooling and Gaussian spatial attention. It strengthens responses in potential target regions and suppresses irrelevant background interference. TGFAENet and GASPPF work together. They form a detection pipeline from spatial perception to structural analysis and then to reasoning-based decision-making.

Experiments on multiple remote sensing object detection datasets show stable performance gains. The proposed method performs better in small target detection, background suppression, and high-precision localization. Visualization results further show that EF-CoT generates more compact target responses. It also reduces diffuse activations in background regions. These results indicate that structured visual reasoning improves detection robustness in complex remote sensing scenes.

The main contributions are summarized as follows.

1) This work introduces the EF-CoT paradigm. The paradigm reformulates detection as a structured reasoning process. Token modeling, spatial priors, and reasoning are unified in one framework.
2) This work builds TGFAENet as the domain-specific tokenizer and reasoning module. Edge-frequency decomposition generates structured tokens. Cross-branch interactions enable token-level reasoning.
3) This work designs the GASPPF module. Dual-branch spatial pyramid pooling aggregates multiscale context. Gaussian spatial attention encodes continuous spatial priors. Feature maps show stronger responses in target regions.
4) Experiments validate the framework on multiple remote sensing datasets. Mean average precision (mAP) increases across benchmarks. Improvements are larger on small-object subsets.

II. Related Work

A. Feature Modeling for Remote Sensing Object Detection

Remote sensing object detection aims to locate and identify objects with specific semantic categories in remote sensing images. Given an input image I, a traditional detector usually learns an end-to-end mapping from the image to detection results: $Y = \mathcal{D}(I)$, where $Y = \{(b_i, c_i, s_i)\}_{i=1}^{N}$ denotes the detection result set. b_i, c_i, and s_i denote the bounding box, the class label, and the confidence score of the ith object, respectively. $\mathcal{D}(\cdot)$ denotes the detection network. Remote sensing images contain large-scale variations, complex background textures, dense object distributions, and many small targets. A direct mapping framework, therefore, often produces unstable boundary localization, missed detections of weak small targets, and false alarms in background regions.

Multiscale feature modeling is a main route for improving remote sensing object detection. Typical methods use feature pyramid structures and fuse features from different levels: $F_l^{out} = \mathcal{F}_{fuse}(F_l, \mathcal{U}(F_{l+1}))$, where F_l denotes the feature at the lth level, $\mathcal{U}(\cdot)$ denotes the upsampling operation, and $\mathcal{F}_{fuse}(\cdot)$ denotes the cross-level feature fusion function. This design improves multiscale object detection through interactions between high-level semantic information and low-level spatial details. Based on this idea, DAE-YOLO and DOMino-YOLO improve detection robustness under complex backgrounds by redesigning feature fusion paths and context suppression mechanisms [22], [23]. Some studies further intro-

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duce multibranch structures or temporal modeling mechanisms to strengthen cross-scale semantic interaction [24].

Frequency-domain modeling has also been used to reduce feature degradation in small targets. In general, an input feature F can be decomposed into a low-frequency structural component and a high-frequency detail component: $F = F_L + F_H$, $F_L = \mathcal{L}(F)$, and $F_H = F - \mathcal{U}(F_L)$, where F_L mainly encodes global contours and semantic structures, while F_H mainly preserves edges, textures, and small-target details. FANet and FKIFM-DETR jointly model low-frequency semantics and high-frequency details. This design reduces the information decay of small targets in deep features [25], [26]. CFBA-FPN enhances the transfer of shallow structural information to high-level semantic spaces through cross-scale frequency compensation [27]. In addition, methods based on region filtering, lightweight network design, and adaptive sampling increase the density of useful information while reducing computational cost [28], [29], [30].

Edge information also benefits remote sensing object detection. In scenes with blurred boundaries, dense targets, or complex background textures, edge constraints improve the geometric separability between objects and backgrounds. A typical edge-guided process can be written as $E = \nabla(F)$ and $F' = F \odot \sigma(\mathcal{G}(E))$, where E denotes the edge response, $\nabla(\cdot)$ denotes a gradient or edge extraction operator, $\sigma(\cdot)$ denotes the sigmoid function, $\odot$ denotes the elementwise multiplication, and $\mathcal{G}(\cdot)$ denotes the edge mapping function. This process usually uses edge information to reweight the original feature. It strengthens responses around object boundaries. Under complex imaging conditions, frequency-adaptive convolution and edge enhancement modules further show the value of frequency information and boundary information for remote sensing object detection [35].

The above methods have improved multiscale feature modeling, frequency-domain enhancement, and edge modeling. However, their core idea still belongs to feature enhancement or feature fusion. Existing methods mainly focus on stronger feature representations. They rarely model intermediate evidence, structural relations, and step-by-step decision paths during detection. When targets are extremely small, boundaries are blurred, or background interference is strong, direct feature fusion may still mix structural information, detailed information, and edge information. This mixing can reduce model stability and interpretability.

B. Degradation- and Frequency-Aware Remote Sensing Modeling

In recent years, remote sensing visual modeling under complex degradation conditions has also received broad attention. Unlike conventional object detection, this line of research mainly improves the usability of remote sensing images from the perspective of image restoration and degradation suppression. However, its designs for structure preservation, frequency-domain constraints, and multiscale interaction also provide useful insights for remote sensing object detection. For example, Wen et al. [36] proposed OBCOT. It formulates unpaired image dehazing as an optimal transport problem. A structure-preserving transport cost constrains the spatial consistency of restored results. Bayesian frequency-domain regularization further aligns the frequency distribution between dehazed images and clear images. These designs improve restoration quality in real degradation scenes [36]. SFTOT further introduces structure preservation, frequency regularization, and text-prompt guidance into unpaired rain streak and raindrop removal. Frequency-domain sparsity constraints and text-guided optimal transport improve structural fidelity and visual realism in restored images [37]. ADSM addresses all-weather degradation restoration. CLIP helps generate latent prompts related to degradation types, degradation attributes, and image content. A wavelet-oriented noise estimating network improves the model perception of different weather degradations and frequency components [38]. For remote sensing image dehazing, CINet models cross-level and intralevel feature relations through cross-level multiscale interaction, intralevel dual-dimension fusion, and multiview progressive extraction. This design improves restoration quality while reducing computational cost [39].

These methods show the value of structure preservation, frequency consistency, degradation-aware prompts, and cross-level interaction in complex remote sensing vision tasks. However, these works mainly focus on image restoration or degradation removal. Their goal is to recover clear visual content from degraded images. They do not directly address category discrimination, boundary localization, or missed detections of small targets in object detection. In contrast, this article does not use frequency information or edge information only as image restoration priors. It explicitly constructs low-frequency structure, high-frequency detail, and edge constraints as structured tokens for reasoning. It also builds a feature-space reasoning path inside the detection network. This path follows evidence extraction, relation modeling, and decision fusion. Therefore, EF-CoT focuses on using edge-frequency information to improve object perception, boundary localization, and decision stability in complex remote sensing scenes.

C. Tokenization in Vision

In recent years, Transformer architectures have shifted visual representation from dense feature-map modeling to token-sequence modeling. Given an image I, a typical Vision Transformer divides it into fixed-size image patches and maps each patch into a token: $T_i = \mathcal{P}(I_i) + e_i$, where I_i denotes the ith image patch, $\mathcal{P}(\cdot)$ denotes the linear projection function, and e_i denotes the positional encoding. The token sequence then models global relations through self-attention: $\mathrm{Attn}(Q, K, V) = \mathrm{Softmax}(QK^\top / \sqrt{d})V$. This mechanism captures long-range dependencies. It has achieved good results in visual classification, detection, and segmentation. Swin Transformer further uses hierarchical window partition and shifted windows. This design balances local modeling and global receptive fields [19].

In object detection, the definition of tokens extends from patch tokens to object query tokens. DETR formulates object detection as a set prediction problem. A set of learnable query vectors represents potential objects: $Y = \mathcal{H}(\mathrm{Decoder}(Q, F))$, where Q denotes the object queries, F denotes the image

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features, and $\mathcal{H}(\cdot)$ denotes the detection head. Deformable DETR further introduces sparse sampling attention. It performs feature interaction only at a few key positions. This design improves convergence speed and detection efficiency [18]. In addition, DynamicViT and TokenLearner reduce computational cost through dynamic token pruning or adaptive token learning. They also make token representations more compact [40], [41].

However, existing visual tokenization methods share one clear limitation. Token construction usually depends on spatial partitioning, query learning, or data-driven compression. It lacks explicit physical semantics and structural constraints. Patch-based tokens mainly represent local image patches. Query-based tokens mainly represent potential object hypotheses. Dynamic tokens mainly target computational efficiency. These tokens can interact globally through attention. Yet, they usually do not explicitly separate low-frequency structure, high-frequency detail, and edge geometry.

This limitation becomes more evident in remote sensing object detection. Remote sensing of small targets often requires high-frequency details for reliable recognition. Object localization in complex backgrounds also relies heavily on edge and structural constraints. If token construction does not explicitly preserve these cues, the following attention layers or detection heads can only learn target cues from mixed features. This process easily causes missed detections of weak targets and boundary localization errors.

Different from the above methods, TGFAENet builds edge-frequency structured tokens with explicit physical semantics. Specifically, the input feature is explicitly decomposed into low-frequency tokens, high-frequency tokens, and edge tokens: $\mathcal{T} = \{T_{lf}, T_{hf}, T_{edge}\}$, where T_{lf} encodes global structure, T_{hf} captures local details, and T_{edge} provides boundary constraints. This design changes the role of tokens. Tokens are no longer only spatial patches or abstract queries. They become intermediate evidence representations with structural, frequency, and geometric semantics. These representations provide a more interpretable basis for the following reasoning process.

D. Reasoning Mechanisms in Vision

Visual reasoning methods aim to model relations among different regions, objects, or semantic components in an image. Early convolutional neural networks mainly aggregate features through local convolution: $F' = \mathrm{Conv}(F)$, where the relation modeling ability is mainly limited by the kernel size and receptive field. Nonlocal neural networks later enhanced long-range feature association through global response modeling. Its general form can be written as $y_i = 1/\mathcal{C}(x)\sum_j f(x_i, x_j)g(x_j)$, where $f(x_i, x_j)$ denotes the relation weight between position i and position j, and $g(\cdot)$ denotes the feature transformation function. Relation networks for object detection further model relations among candidate objects in object detection. Graph-based methods, such as Graph R-CNN, model regions or objects as nodes and perform high-level semantic interaction through graph propagation: $H^{(k+1)} = \sigma(AH^{(k)}W^{(k)})$, where A denotes the relation adjacency matrix, $H^{(k)}$ denotes node features at the kth layer, and $W^{(k)}$ denotes the learnable parameter matrix [42], [43], [44]. These methods improve relational modeling ability. However, their reasoning processes are still mostly implicit. In other words, attention, nonlocal operations, or graph propagation enable feature interaction. Yet, they usually do not explicitly define staged reasoning states, such as evidence extraction, relation modeling, and decision generation. The detection result can still be summarized as a direct mapping from features to outputs: $Y = \mathcal{D}(F)$.

In recent years, CoT has improved complex problem-solving in natural language processing and multimodal tasks through intermediate reasoning steps. Its core idea reformulates direct output generation as a staged reasoning process: $Z_1 = \Phi_1(X)$, $Z_2 = \Phi_2(Z_1)$, $Y = \Phi_3(Z_2)$, where Z_1 and Z_2 denote intermediate reasoning states. The final output Y is generated from progressively updated intermediate evidence. This idea provides useful guidance for complex visual tasks. A detection model should not rely only on a one-step mapping from features to outputs. It should explicitly model intermediate evidence and structural relations.

EF-CoT introduces this idea into the feature space of remote sensing object detection. EF-CoT is not a language-form CoT. It also does not simply rename a multilayer network as a reasoning process. Its key design is to explicitly construct intermediate token states with physical semantics. Staged operators then build a reasoning path from structural perception to relation modeling and then to decision fusion: $Z_0 = \{T_{lf}, T_{hf}, T_{edge}\}$, $Z_1 = \Phi_{ext}(Z_0)$, $Z_2 = \Phi_{rel}(Z_1)$, and $Y = \Phi_{dec}(Z_2)$, Here, Z_0 denotes the initial evidence set formed by edge-frequency tokens. Z_1 denotes the target response state after evidence extraction. Z_2 denotes the structural consistency state after relation modeling. Y denotes the final detection result. Compared with traditional visual reasoning methods, EF-CoT uses a different reasoning object. It does not reason over general region features or abstract object queries. It reasons over edge-frequency tokens with explicit structure, detail, and boundary semantics. Its reasoning process is also different. It is not a single attention interaction. It is a staged feature-space reasoning process built around intermediate states.

III. Method

A. EF-CoT Paradigm

Traditional object detectors usually formulate detection as a direct mapping from features to predictions. Their intermediate feature states are not explicitly defined. This article reformulates remote sensing object detection as a differentiable reasoning process in the feature space. We term this process EF-CoT. Unlike language-based CoT, EF-CoT does not generate textual rationales. It performs step-by-step reasoning in the feature space with structured visual tokens. Given an input feature map $X \in \mathbb{R}^{H\times W\times C}$, EF-CoT first constructs an edge-frequency structured token set

$$Z^{(0)} = \mathcal{T}(X) = \{T_{lf}, T_{hf}, T_{edge}\} \tag{1}$$

where T_{lf}, T_{hf}, and T_{edge} denote low-frequency structural tokens, high-frequency detail tokens, and edge-aware boundary tokens, respectively. These tokens form the initial

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reasoning state. The following reasoning process is defined as a sequence of state transitions:

$$Z^{(1)} = \Phi_{ext}\left(Z^{(0)}\right) \tag{2}$$

$$Z^{(2)} = \Phi_{rel}\left(Z^{(1)}\right) \tag{3}$$

$$Y = \Phi_{dec}\left(Z^{(2)}\right) \tag{4}$$

where Φ_{ext}, Φ_{rel}, and Φ_{dec} correspond to evidence extraction, relation modeling, and decision fusion, respectively. $Z^{(1)}$ is the evidence state. It enhances target-related responses and suppresses background interference. $Z^{(2)}$ is the structural relation state. It models interactions among structural cues, detail cues, and boundary cues. The final prediction Y is generated from the decision state. It is not produced directly from the original feature map.

This formulation clarifies the meaning of "reasoning" in EF-CoT. The model does not simply stack multiple feature processing layers. It explicitly defines token types, intermediate reasoning states, and state-transition operators. Each state has a specific functional role. This design builds a traceable reasoning path from structured perception to final detection.

B. Edge-Frequency Structured Tokenizer

The tokenizer aims to provide structured visual evidence for the following reasoning process. Remote sensing objects usually contain both global shape cues and local boundary details. Therefore, this article decomposes the input feature map into three components: low-frequency, high-frequency, and edge-aware components

$$T_{lf} = \text{DownUp}(X) \tag{5}$$

$$\text{T}_{hf} = \text{X-T}_{lf} \tag{6}$$

$$T_{edge} = |\nabla_x X| + |\nabla_y X|. \tag{7}$$

The low-frequency token T_{lf} captures global structural information and suppresses local noise. The high-frequency token T_{hf} preserves fine-grained details. These details are especially important for small targets. The edge token T_{edge} provides boundary-aware geometric constraints for localization. Compared with ordinary patch tokens or implicit feature maps, these tokens have clearer physical semantics. They are more suitable for step-by-step reasoning.

The tokenizer differs from existing frequency-aware or edge-guided modules in its design goal. Existing methods usually use frequency components or edge responses as enhancement terms. They reweight, compensate, or concatenate the original features. Their goal is to obtain stronger fused features. In contrast, this tokenizer does not directly enhance features. It constructs reasoning units with clear semantic roles. Low-frequency tokens, high-frequency tokens, and edge tokens correspond to three types of detection evidence: structure, detail, and boundary. They play different roles in the following reasoning stages.

Specifically, low-frequency tokens provide global structural context. They reduce semantic confusion under complex backgrounds. High-frequency tokens preserve small targets and local texture details. They reduce small-target degradation in deep features. Edge tokens serve as geometric constraints and modulate high-frequency responses. They improve boundary consistency around targets. The three types of tokens jointly form the initial evidence set. EF-CoT then updates them through staged operators. As a result, the detection process changes from "feature enhancement followed by direct prediction" to "structured evidence construction, relation modeling, and decision fusion." This design is the main difference between the proposed tokenizer and existing frequency-aware or edge-guided methods.

C. Differentiable Token-Level Reasoning and Training

After tokenization, EF-CoT performs reasoning through three stages. First, the evidence extraction operator Φ_{ext} enhances discriminative token responses and generates the evidence state $Z^{(1)}$. Second, the relation modeling operator Φ_{rel} models interactions among low-frequency, high-frequency, and edge-aware tokens. It generates the structural relation state $Z^{(2)}$. Finally, the decision fusion operator Φ_{dec} aggregates the refined token representations and feeds them into the detection head.

During training, EF-CoT is implemented as a fully differentiable neural network process. This process does not introduce external symbolic reasoning, handcrafted rules, or postprocessing operations. The model is optimized end-to-end with the standard detection loss

$$\mathcal{L}_{det} = \mathcal{L}_{cls} + \mathcal{L}_{box} + \mathcal{L}_{dfl}. \tag{8}$$

Gradients are back-propagated from the detection head. They pass through Φ_{ext}, Φ_{rel}, Φ_{dec}, and the tokenizer $\mathcal{T}(\cdot)$ in sequence. Therefore, the intermediate reasoning states are learned together with the detector. No extra annotations or auxiliary supervision are required.

EF-CoT differs from a traditional sequential feature processing pipeline in two aspects. First, a standard sequential pipeline mainly performs layer-by-layer feature transformation. Its intermediate features usually do not have predefined semantic roles. EF-CoT explicitly defines the reasoning states $Z^{(0)}$, $Z^{(1)}$, and $Z^{(2)}$. They correspond to structured tokenization, evidence extraction, and relation modeling, respectively. Second, standard feature fusion usually mixes different cues directly. EF-CoT instead refines token representations step by step before prediction. This design avoids premature mixing of structure, texture, and boundary information. It improves localization stability in complex remote sensing scenes.

D. TGFAENet: Instantiation of EF-CoT

The EF-CoT paradigm models detection as a CoT process over structured tokens. The formulation requires a concrete network to implement the mapping from $\mathcal{T}$ to Y. Fig. 3 shows the proposed TGFAENet. The network serves as a structured instantiation of EF-CoT.

From a formal perspective, TGFAENet implements the following mapping:

$$\mathcal{F}_{\text{TGFAE}} : \mathcal{T} \to Y. \tag{9}$$

This mapping satisfies

$$\mathcal{F}_{\text{TGFAE}} \approx \Phi_{\text{dec}} \circ \Phi_{\text{rel}} \circ \Phi_{\text{ext}}. \tag{10}$$

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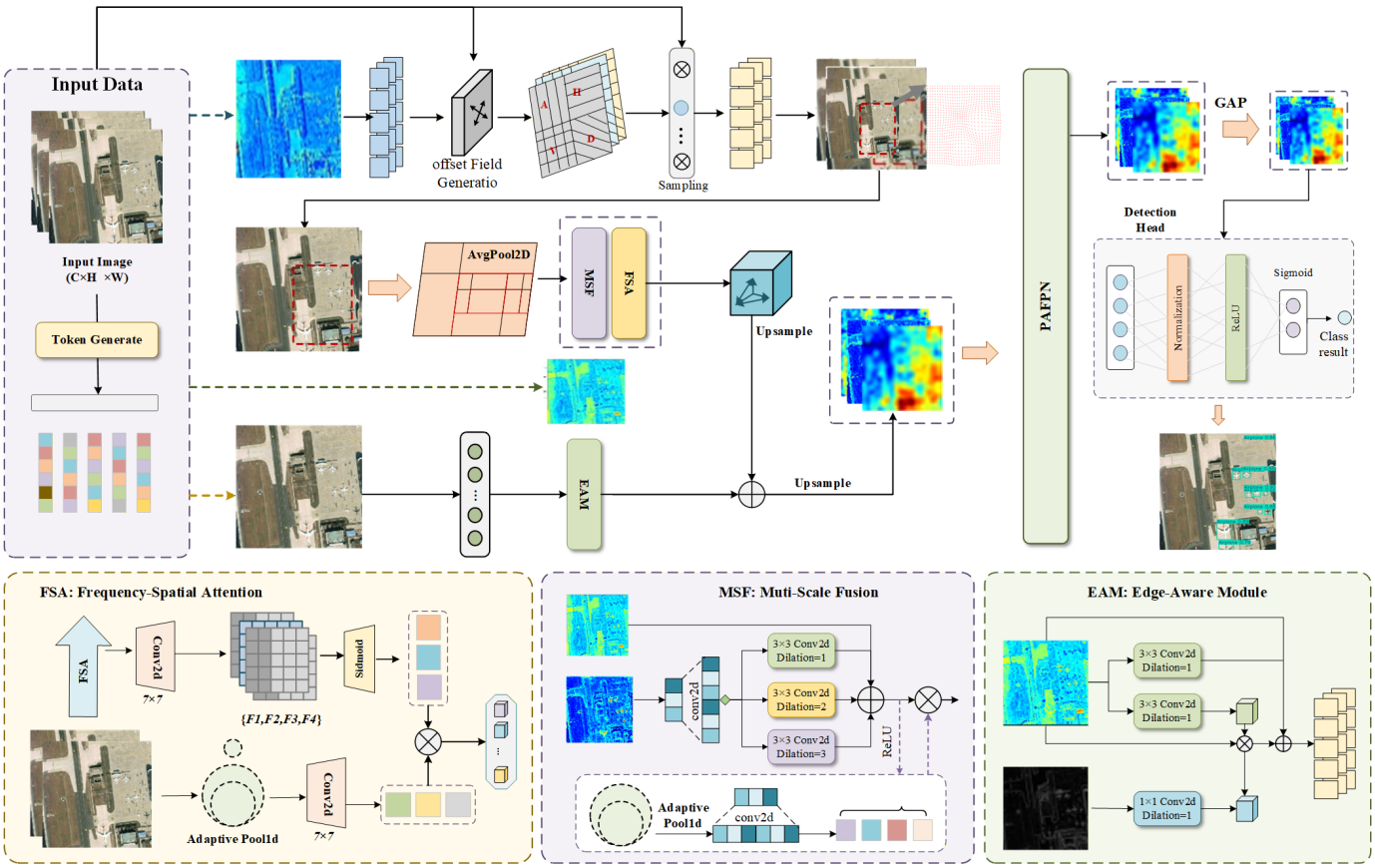

Fig. 3. Overall architecture of the proposed TGFAENet. Under the EF-CoT paradigm, object detection is formulated as a CoT reasoning process based on structured tokens. To realize the inference path from input features to final predictions, TGFAENet is designed as a structural instantiation of this paradigm, explicitly modeling feature evolution and information interaction for progressive reasoning and decision refinement.

1) Frequency-Guided Spatial Reallocation: TGFAENet departs from direct feature extraction in the original space. A spatial redistribution mechanism is introduced using high-frequency tokens. The high-frequency component T_{hf} generates an offset field

$$\Delta \mathbf{p} = \mathcal{O}(T_{\text{hf}}) \tag{11}$$

where $\mathcal{O}(\cdot)$ denotes a convolution mapping. A deformable sampling grid is then constructed

$$\mathbf{G} = \mathbf{G}_0 + \Delta \mathbf{p} \tag{12}$$

where $\mathbf{G}_0$ represents the regular grid. Based on this grid, input features are resampled

$$I' = \mathcal{S}(I, \mathbf{G}) \tag{13}$$

where $\mathcal{S}(\cdot)$ denotes the bilinear interpolation sampling. This process converts uniform sampling into frequency-guided nonuniform sampling. Sampling density increases in high-frequency regions.

Fig. 4 illustrates this mechanism. High-frequency components highlight boundary and structural cues. The model generates spatial offsets from these responses. The regular grid is deformed accordingly. Features are sampled at shifted locations.

Sampling becomes content-adaptive. Dense sampling appears in target regions. Sparse sampling remains in background regions. Redundant computation is reduced. Representation of fine details is strengthened.

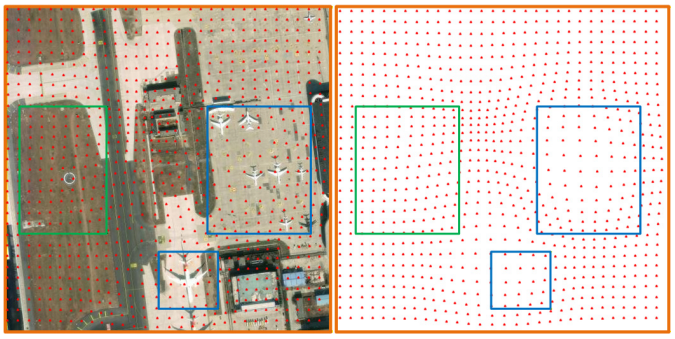

Fig. 4. Illustration of the deformable sampling grid. TGFAENet adaptively generates offsets from high-frequency cues to deform the regular grid and resample features. Blue regions indicate dense sampling on objects, while green regions denote sparse sampling on the background, achieving frequency-guided nonuniform sampling.

2) Triple-Branch Token Reasoning: After spatial redistribution, TGFAENet performs parallel reasoning in three subspaces. The low-frequency branch is $\tilde{T}_{\text{lf}} = \Phi_{\text{lf}}(T'_{\text{lf}})$. The high-frequency branch is $\tilde{T}_{\text{hf}} = \Phi_{\text{hf}}(T_{\text{hf}}, T_{\text{edge}})$. The low-frequency component T'_{lf} is recomputed from the resampled features I'. The high-frequency branch applies an edge-guided

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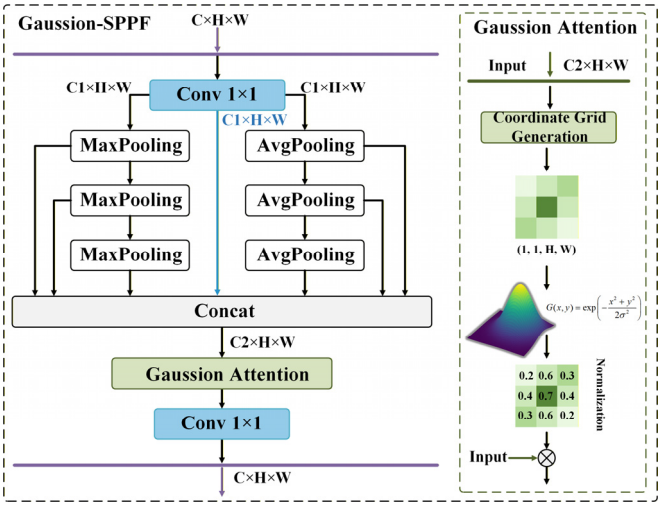

Fig. 5. Overall architecture of the proposed GASPPF.

modulation mechanism

$$\tilde{T}_{\mathrm{hf}} = T_{\mathrm{hf}} \cdot \left(1 + \sigma\left(\mathcal{M}\left(T_{\mathrm{hf}}\right)\right) \cdot \sigma\left(\mathcal{E}\left(T_{\mathrm{edge}}\right)\right)\right) + \mathcal{M}\left(T_{\mathrm{hf}}\right) \quad (14)$$

where $\sigma(\cdot)$ denotes the sigmoid function, $\mathcal{M}(\cdot)$ denotes a 3×3 convolution block, and $\mathcal{E}(\cdot)$ denotes a 3×3 convolution block. This formulation shows that the edge token T_{edge} participates through multiplicative modulation. Boundary responses are strengthened during high-frequency reconstruction. After reasoning, branch outputs are progressively fused

$$Y = \mathcal{H}\left(\left[\tilde{T}_{\mathrm{lf}} \uparrow, \tilde{T}_{\mathrm{hf}}\right]\right) \quad (15)$$

where $\uparrow$ denotes the upsampling operation. This fusion is not a direct concatenation. Structural information is integrated first. Detailed information is added step by step. The final output preserves global consistency and local precision.

TGFAENet combines frequency-guided resampling, three-branch reasoning, and progressive fusion. This design converts the abstract EF-CoT process into a computable structure. The mapping $\mathcal{T} \rightarrow Y$ is explicitly implemented.

E. GA Spatial Pyramid Pooling Fusion

Although TGFAENet performs structured reasoning at the token level, it mainly focuses on interactions among edge, frequency, and structural cues. It does not explicitly model the continuity of spatial responses. Therefore, as shown in Fig. 5, this article further introduces GASPPF. GASPPF injects a smooth spatial prior at the feature level. It improves the model's perception of potential target regions.

First, dual-branch spatial pyramid pooling constructs multiscale contextual representations

$$F_{\mathrm{ms}} = \mathcal{P}_{\max}\left(X\right) \oplus \mathcal{P}_{\mathrm{avg}}\left(X\right) \quad (16)$$

where $\oplus$ denotes the multiscale concatenation. This process aggregates contextual information under different receptive-field scales. It improves feature representation for multiscale targets and complex backgrounds. After multiscale features are obtained, Gaussian attention (GA) is introduced to continuously modulate spatial responses

$$A\left(x, y\right) = \exp\left(-\frac{(x - x_0)^2 + (y - y_0)^2}{2\sigma^2}\right). \quad (17)$$

The Gaussian weight is then applied to the feature map

$$\hat{F}\left(x, y\right) = F_{\mathrm{ms}}\left(x, y\right) \cdot A\left(x, y\right). \quad (18)$$

GA does not assume that remote sensing targets always appear near the image center. Targets in remote sensing images may appear at arbitrary locations. A fixed center-based localization assumption is, therefore, unreasonable. In this article, GA acts as a weak spatial regularization term. It provides a smooth and continuous spatial confidence distribution for the feature map. It does not determine the object's location. The final object location is still predicted jointly by multiscale feature responses, edge-frequency token reasoning, and the detection head.

This center-decay form is still reasonable for remote sensing detection. First, remote sensing images are usually cropped or resized into fixed-size inputs during training and inference. Image borders are more likely to contain truncated objects, incomplete context, crop boundaries, and stitching artifacts. These factors produce unstable responses. In contrast, the center region and its neighborhood usually retain more complete contextual information. A higher base confidence around this region can suppress false responses caused by border truncation and boundary noise.

Second, GA uses continuous decay. It does not hard-mask border regions. Real targets located away from the image center can still be detected through strong semantic responses, edge responses, and high-frequency details. In other words, this module does not exclude border targets. It only reduces the interference of spatially unreliable noise in the following reasoning process.

From the EF-CoT perspective, GASPPF can be regarded as spatial reliability modulation in the token space

$$\mathcal{T}' = \mathcal{T} \odot A \quad (19)$$

where A denotes the edge-frequency structured tokens constructed by TGFAENet and G denotes the Gaussian spatial weight. Through this modulation, spatially stable regions obtain higher responses. Unreliable responses caused by crop boundaries, shadows, background textures, or local noise are relatively suppressed. This process provides smoother and more stable inputs for evidence extraction, relation modeling, and decision fusion.

Therefore, GA in GASPPF is not a strong assumption about object location. It is a weak prior constraint for improving spatial continuity and suppressing boundary noise. It complements the edge-frequency structured reasoning in TGFAENet. GASPPF provides smooth spatial reliability constraints. TGFAENet uses low-frequency structure, high-frequency details, and edge information for object localization and decision-making. This design improves feature stability and detection robustness in complex remote sensing scenes without restricting targets to fixed spatial locations.

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F. Overall Pipeline

Combining all modules, the proposed method defines a complete inference pipeline from input to output. Given an input image X, the overall process is formulated as follows:

$$Y = \mathcal{F}_{\text{GASPPF}} \circ \mathcal{F}_{\text{TGFAE}} \circ \mathcal{G}(I) \tag{20}$$

where $\mathcal{G}(\cdot)$ denotes the tokenizer. $\mathcal{F}_{\text{TGFAE}}$ denotes the token reasoning network. $\mathcal{F}_{\text{GASPPF}}$ denotes the spatial prior enhancement module. The EF-CoT framework is further written in a unified form

$$Y = \mathcal{F}(I) = \mathcal{F}_{\text{prior}}(\Phi_{\text{CoT}}(\mathcal{G}(I))) \tag{21}$$

where $\Phi_{\text{CoT}}(\cdot)$ denotes the CoT reasoning operator. Compared with conventional methods $Y = \mathcal{F}(I)$, this work implements a decomposed reasoning pipeline $I \rightarrow \mathcal{T} \rightarrow$ Reasoning $\rightarrow$ Prior-guided Fusion $\rightarrow Y$.

IV. Experiments and Analysis

A. Dataset Description

To evaluate the effectiveness, robustness, and cross-dataset generalization of the proposed method in complex remote sensing scenes, this article uses four representative public remote sensing object detection datasets: RS-STOD, RSOD, NWPU VHR-10, and DIOR. These datasets differ in imaging platform, spatial resolution, sample size, scene complexity, and annotation format. They evaluate the model from several aspects, including dense small-object detection, multiscale object recognition, category diversity modeling, and large-scale scene generalization. Table I reports the category definitions, sample sizes, training/validation/test splits, and annotation information of these datasets [45], [46], [47].

1) RS-STOD is a public dataset for small-object detection in remote sensing images. Its images mainly come from UAV aerial imagery and high-resolution remote sensing images. This dataset contains extremely small objects and dense object distributions. Its backgrounds also contain many structures that look similar to real targets, such as road textures, building edges, shadow regions, and high-frequency background noise. These factors can bury small-target responses in background responses. Missed detections and false positives can then occur. Therefore, RS-STOD mainly evaluates fine-grained discrimination, weak-target response modeling, and background-interference suppression under the dense small-object regime. Experiments on this dataset examine whether EF-CoT can reduce small-target feature degradation by jointly modeling low-frequency structure, high-frequency detail, and edge information. This dataset contains five categories: small vehicle (C1), large vehicle (C2), ship (C3), airplane (C4), and storage tank (C5).
2) RSOD is a classical benchmark dataset widely used in remote sensing object detection. It contains several typical remote sensing object categories. Compared with RS-STOD, RSOD has a more diverse object scale distribution. It contains both small objects and medium-scale objects. It also covers multiple remote sensing scenes, such as airports, oil tank areas, roads, bridges, and playgrounds. Therefore, this dataset is suitable for evaluating feature representation and cross-scene adaptation under multiscale object detection settings. For remote sensing detectors, the challenges in RSOD come not only from scale variation. They also come from appearance differences among categories and structural confusion in complex backgrounds. For example, overpass shares a strong structural similarity with road backgrounds. Oil tank instances can show boundary adhesion when they are densely distributed. This article uses RSOD to verify detection stability and boundary localization ability when multiscale objects coexist. This dataset contains four categories: aircraft (C1), oil tank (C2), overpass (C3), and playground (C4).
3) NWPU VHR-10 is a typical high-resolution remote sensing object detection dataset. It contains ten common ground-object categories. Its image sources are diverse to some extent, and its scenes cover a wide range of conditions. Objects in this dataset show large differences in scale, shape, texture, and spatial distribution. Some categories also have strong appearance similarity. For example, a storage tank can look similar to some circular building structures. Harbor and ship often appear close to each other. The bridge has structural coupling with the road or river backgrounds. These factors make NWPU VHR-10 a useful benchmark for evaluating category discrimination, structural modeling, and cross-domain generalization. Experiments on this dataset mainly verify the stability of EF-CoT under category diversity, complex geometry, and multisource imaging conditions. This dataset contains ten categories: airplane (C1), ship (C2), storage tank (C3), baseball diamond (C4), tennis court (C5), basketball court (C6), ground track field (C7), harbor (C8), bridge (C9), and vehicle (C10).
4) DIOR is a large-scale optical remote sensing object detection benchmark. It contains many high-resolution remote sensing images and rich ground-object categories. DIOR has many categories, a wide object-scale range, complex scene types, strong background interference, and clear interclass appearance differences. Compared with RS-STOD, RSOD, and NWPU VHR-10, DIOR has a larger data scale, broader category coverage, and richer scene diversity. It can better test model generalization in large-scale complex remote sensing scenes. The objects in DIOR include typical small or densely distributed targets, such as airplane, ship, and vehicle. They also include large structural targets, such as airport, bridge, dam, stadium, and ground track field. Therefore, DIOR evaluates small-target detail modeling, large-object structure modeling, and multicategory object detection under complex backgrounds at the same time.

This article focuses on horizontal bounding-box detection in remote sensing images. The horizontal annotations in DIOR are consistent with the proposed detection framework. No conversion from rotated boxes to horizontal boxes is required in the experiments. All models are compared under the same

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input size, training strategy, and evaluation metrics. DIOR contains 20 categories: airplane (1), airport (2), baseball field (3), basketball court (4), bridge (5), chimney (6), dam (7), expressway service area (8), expressway toll station (9), golf field (10), ground track field (11), harbor (12), overpass (13), ship (14), stadium (15), storage tank (16), tennis court (17), train station (18), vehicle (19), and windmill (20). This dataset mainly verifies the robustness and generalization of the proposed method in large-scale, multicategory, and complex remote sensing scenes.

Overall, the four datasets play clear and complementary roles in the experimental evaluation. RS-STOD focuses on extremely small targets and dense object distributions. RSOD evaluates feature representation and cross-scene adaptation under multiscale object coexistence. NWPU VHR-10 emphasizes category diversity, structural complexity, and cross-domain stability. DIOR further provides large-scale validation under multicategory and complex-scene conditions. Experiments on these four datasets evaluate EF-CoT across different remote sensing scenes, object scales, and complexity levels. The results provide stronger support for the experimental conclusions.

B. Evaluation Metrics

To comprehensively evaluate the performance of the proposed object detection model, the following metrics are adopted.

1) Precision (P) measures the proportion of correctly predicted positive samples among all positive predictions. It reflects the model's ability to reduce false positives and is defined as follows:

$$Precision = \frac{TP}{TP + FP} \tag{22}$$

where TP refers to targets accurately detected by the model, while FP indicates background areas wrongly predicted as objects.

2) Recall (R) evaluates the proportion of correctly predicted positive samples among all actual positive samples, indicating the model's sensitivity to detecting objects

$$Recall = \frac{TP}{TP + FN} \tag{23}$$

where FN represents the objects missed by the detector, which are mistakenly classified as background.

3) Average precision (AP) quantifies the area under the precision–recall (P–R) curve for *a* specific class and IoU threshold. It provides *a* balanced evaluation of detection accuracy

$$AP_n = \int_0^1 P(x)\,dx \tag{24}$$

where n denotes the number of categories and $P(x)$ represents the P–R curve for category AP. The area under this curve, obtained via integration, corresponds to the AP value for that category.

4) mAP is the average AP across all classes and multiple IoU thresholds

$$mAP = \frac{1}{N}\sum_{i-1}^{N} AP_n \tag{25}$$

where N denotes the total number of categories and AP_n represents the AP value for category n.

5) *FPS:* To evaluate the real-time performance of the detector, FPS is used as *a* key metric. FPS indicates the number of images the detector can process per second, providing an assessment of its inference speed. *A* higher FPS value signifies improved real-time performance of the detector

$$FPS = \frac{1}{T_m} \tag{26}$$

where T_m denotes the time taken by the detector to process a single image.

6) *Parameters and Computational Complexity:* Params and GFLOPs are important metrics for evaluating the lightweight nature of *a* model. Params refers to the number of parameters in the model; the larger the parameter count, the heavier the model weights. GFLOPs indicates the number of floating-point operations required by the model, and higher GFLOPs imply greater computational cost and slower inference speed. Taking the *a* convolution operation with the *a* kernel size of $i \times i$ as an example, the definitions of Params and GFLOPs are given as follows:

$$Params = C_{out} \times \left(k_{i\times i}^2 \times C_{in} + 1\right) \tag{27}$$

$$GFLOPs = \frac{2 \times C_{in} \times k_{i\times i}^2 \times C_{out} \times W_i \times H_i}{10^9} \tag{28}$$

where C_{in} denotes the number of input channels, $k_{i\times i}$ represents a convolution with kernel size $i \times i$, C_{out} denotes the number of output channels, and W_i and H_i represent the width i and height i of the feature map, respectively.

C. Training Strategies and Implementation Details

To ensure comparability and reproducibility, all models are trained and evaluated under a unified software and hardware environment. The platform uses an Intel i9-13900K CPU and an NVIDIA RTX 3090 GPU (24 GB). Implementation is based on Python 3.8 and PyTorch 2.2.2, with CUDA 11.8 acceleration. Identical environments remove performance variance caused by hardware differences. The comparison remains consistent across all methods [42].

All methods follow an end-to-end optimization strategy. The initial learning rate is set to 0.001. Linear warm-up stabilizes early training. Cosine annealing ensures smooth convergence in later stages. The optimizer is Adam. The momentum parameter is set to 0.937. The weight decay is set to 5×10^{-4}. All models are trained for 200 epochs. The batch size is fixed at 8.

Input images are resized to 640×640. This setting reduces scale variance across datasets. During inference, a unified post-processing configuration is applied. The confidence threshold

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TABLE I
DETAILS OF RS-STOD, RSOD, NWPU VHR-10, AND DIOR DATASETS

Dataset	Class type	Total images	Train/Val/Test	Annotation	Resolution	Main Challenge
RS-STOD	5	2315	1852/231/232	HBB	512×512	Dense small objects
RSOD	4	936	655/93/188	HBB	1044×915	Multi-scale objects
NWPU VHR-10	10	800	560/80/160	HBB	Variable	Class diversity
DIOR	20	23463	5862/5863/11738	HBB	800×800	Large-scale complex scenes

is set to 0.25. The IoU threshold for nonmaximum suppression is set to 0.45. Redundant predictions are suppressed while maintaining recall.

All training and inference settings remain identical across RS-STOD, RSOD, NWPU VHR-10, and DIOR. This controlled setup isolates the effect of model design. Performance differences can be attributed to architectural changes rather than training strategies.

D. Cross-Dataset Evaluation on Remote Sensing Benchmarks

1) Baseline Selection: YOLOv11n is selected as the unified baseline. Reported results show a balance between detection accuracy and inference speed. The model meets both real-time and performance requirements in deployment settings. This behavior makes it suitable as a reference model.

From a structural view, YOLOv11n follows a standard CNN-based pipeline. Multiscale feature fusion is integrated into the framework. The overall design remains consistent with typical detection architectures. This property allows controlled modifications. Structural extensions and module replacements can be evaluated without introducing additional variables. Performance changes can be attributed to representation improvements.

Training logs show stable convergence behavior. The architecture remains simple. These properties support systematic comparisons and ablation studies. Variance across runs is reduced. Experimental conclusions become easier to interpret. Based on these observations, YOLOv11n is adopted as the unified baseline to ensure fair and consistent evaluation.

2) Overall Performance Comparison: Cross-dataset experiments are conducted on RS-STOD, RSOD, NWPU VHR-10, and DIOR. These datasets differ in object scale, scene complexity, and category diversity. This setup evaluates detection ability and generalization from multiple perspectives.

Comparison results show unstable performance for existing methods across datasets. Performance drops appear in dense small-object scenes. Larger drops appear under strong background interference.

The proposed method maintains consistent leading performance on all four datasets. This behavior indicates stronger cross-scene robustness and stability. On RS-STOD (see Fig. 6), results show higher accuracy under extremely small objects and dense distributions. Detection outputs contain fewer false positives in cluttered regions. This behavior reflects a better fine-grained representation and interference suppression. On RSOD (see Fig. 7), results show stable performance under multiscale object coexistence. Detection accuracy remains high across different object sizes. This observation verifies effective multiscale modeling. On NWPU VHR-10 (see Fig. 8), results remain stable under diverse categories and cross-scene variations. Predictions show consistent quality across different domains. This behavior indicates improved cross-domain generalization.

Table II reports comprehensive quantitative results. EF-CoT achieves the best performance on three datasets. Results reach 88.6%/64.5% (mAP50/mAP50-95) on RS-STOD. Results reach 97.8%/72.5% on RSOD. Results reach 96.5%/67.2% on NWPU VHR-10. These values exceed YOLOv8n and Hyper-YOLO-T.

Improvements are larger on mAP50-95. High-IoU regions show better localization quality. Bounding boxes align more accurately with object boundaries.

Efficiency metrics show competitive performance. The model uses 3.28M parameters. Computation cost is 7.5 GFLOPs. Inference speed reaches 228 FPS. These values indicate a balance between accuracy and efficiency. Compared with RT-DETR, the computation cost is lower while maintaining high accuracy. Deployment feasibility is improved.

From the perspective of runtime cost, EF-CoT achieves an average inference latency of 4.39 ms/image, corresponding to 228 FPS. Its speed is slightly lower than that of lightweight detectors, such as YOLOv5n, YOLOv7-Tiny, YOLOv8n, and YOLOv11n. However, EF-CoT still supports real-time inference. EF-CoT also introduces only 3.28M parameters and 7.5 GFLOPs. Its computational cost is much lower than that of Mamba-YOLO-T and RT-DETR. Compared with YOLOv11n, EF-CoT increases the latency from 3.80 to 4.39 ms. This additional 0.59 ms brings mAP50 gains of 15.1%, 3.9%, and 6.1% on RS-STOD, RSOD, and NWPU VHR-10, respectively. These results show that a small runtime cost leads to clear accuracy gains. EF-CoT, therefore, achieves a favorable balance among accuracy, computational cost, and real-time speed. It also shows potential for practical remote sensing detection deployment.

As shown in Table III, EF-CoT achieves the best overall performance on the DIOR dataset. Its mAP50 reaches 78.4%. This result exceeds YOLOv5n, YOLOv8n, YOLOv11n, and Hyper-YOLO-T by 8.2, 4.7, 5.9, and 3.4 percentage points, respectively. EF-CoT also outperforms existing remote sensing detectors, including HAE-Net, CANet, SESA-Net, and MSNet. This result further verifies its effectiveness in large-scale complex remote sensing scenes. Classwise results show that EF-CoT obtains high accuracy on Bridge, Chimney, Overpass, Storage Tank, and Train Station. These categories usually contain complex structures. The results indicate that edge-frequency structured tokens enhance the representation of structurally complex objects. Some methods perform well on specific categories but show unstable overall performance.

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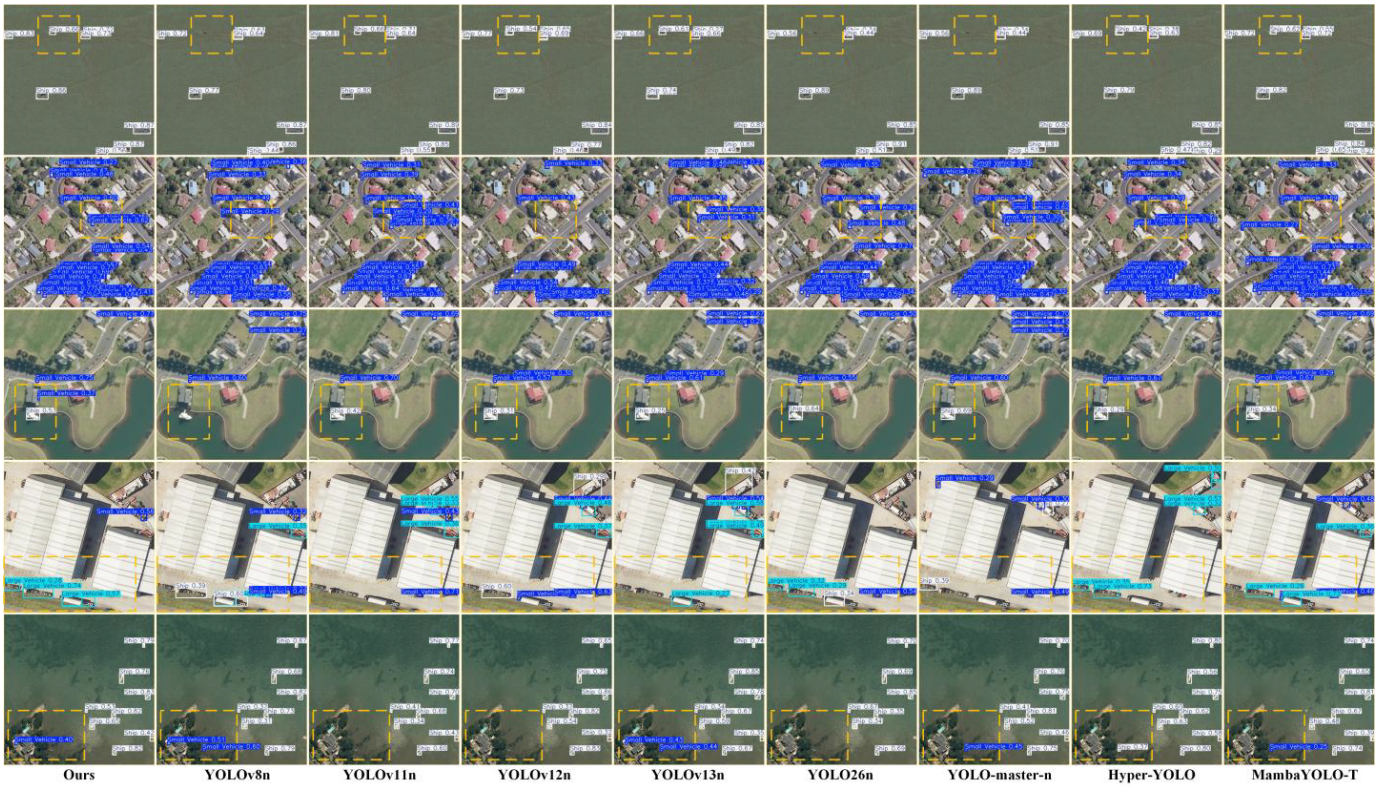

Fig. 6. Performance comparison of different detectors on the RS-STOD dataset. The orange dashed box marks the region where the proposed method achieves higher detection accuracy. This result shows that EF-CoT delivers better overall detection performance than the other methods on this dataset.

Fig. 7. Performance comparison of different detectors on the RSOD dataset. The orange dashed box marks the region where the proposed method achieves higher detection accuracy. This result shows that EF-CoT delivers better overall detection performance than the other methods on this dataset.

EF-CoT maintains balanced detection results across most categories. This pattern shows that the proposed chain reasoning mechanism improves generalization to multiscale, multicategory, and complex-background targets.

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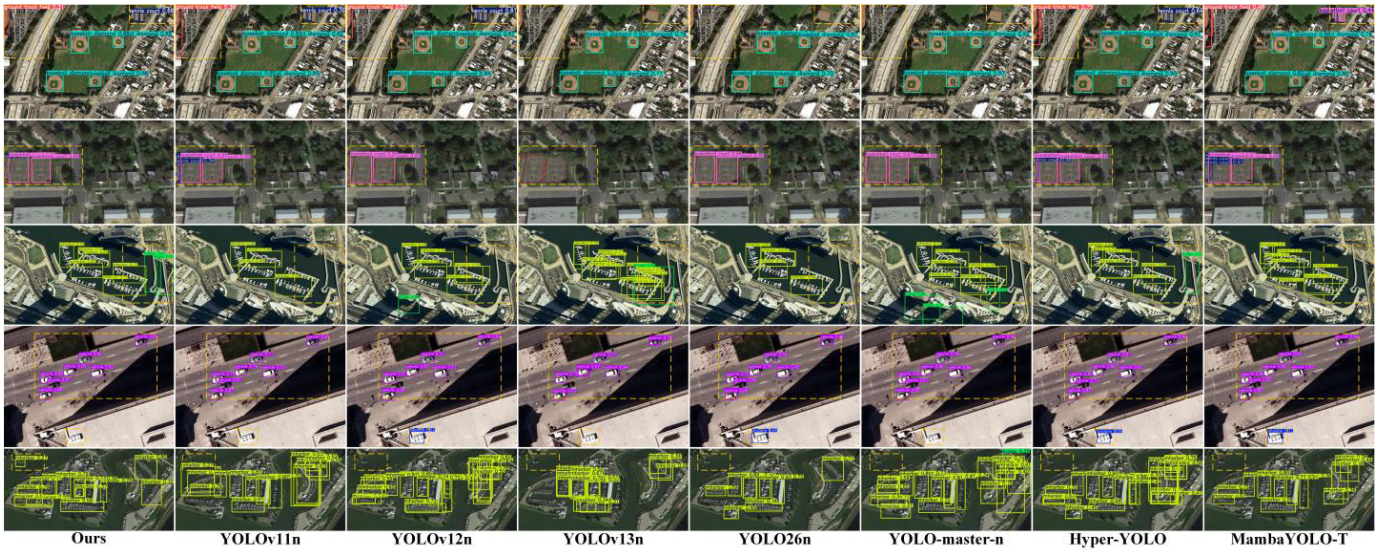

Fig. 8. Performance comparison of different detectors on the NWPU VHR-10 dataset. The orange dashed box marks the region where the proposed method achieves higher detection accuracy. This result shows that EF-CoT delivers better overall detection performance than the other methods on this dataset.

TABLE II

CROSS-DATASET EVALUATION DEMONSTRATING CONSISTENT PERFORMANCE GAINS ACROSS DIVERSE REMOTE SENSING SCENARIOS. THE BEST RESULTS ARE MARKED IN BOLD. UNDERLINED VALUES INDICATE DATA DIRECTLY QUOTED FROM THE ORIGINAL SOURCE. LATENCY DENOTES THE AVERAGE SINGLE-IMAGE INFERENCE TIME, CALCULATED AS 1000/FPS MS UNDER THE SAME TESTING ENVIRONMENT. "–" INDICATES THAT THE CORRESPONDING VALUE IS NOT REPORTED IN THE ORIGINAL PAPER OR CANNOT BE FAIRLY MEASURED UNDER THE SAME SETTING

Method	RS-STOD		RSOD		NWPU VHR-10		Computational overhead and speed			
	mAP_{50}↑	mAP_{50-95}↑	mAP_{50}↑	mAP_{50-95}↑	mAP_{50}↑	mAP_{50-95}↑	Params↓	GFLOPs↓	FPS↑	Latency↓
YOLOv5n	70.2	40.9	91.7	56.8	91.5	54.8	**1.76**	7.1	284	3.52
YOLOv7-Tiny	72.2	41.0	**95.5**	58.7	93.2	57.9	6.01	13.3	**335**	**2.99**
YOLOv8n	75.6	48.9	93.5	61.7	89.5	57.2	3.00	8.1	297	3.37
YOLOv10n	69.8	45.3	90.8	60.8	83.7	53.3	2.69	6.5	218	4.59
YOLOv11n	73.5±0.3	47.5±0.4	93.9±0.2	62.4±0.3	90.4±0.3	57.8±0.4	2.58	6.3	263	3.80
YOLOv12n	71.5	45.4	90.5	60.2	86.7	55.3	2.55	6.3	165	6.06
YOLOv13n	69.6	43.5	72.7	50.5	66.1	47.9	2.44	**6.2**	166	6.02
YOLO26n	70.3	43.6	92.6	59.1	85.8	52.4	2.37	5.2	173	5.78
YOLO-master	72.5	45.9	91.8	61.5	89.7	56.7	7.50	9.6	154	6.49
Hyper-YOLO-T	75.6	49.1	92.2	61.8	90.5	58.1	2.68	7.6	243	4.12
Mamba-YOLO-T	74.8	48.1	92.0	59.7	90.3	55.5	5.66	12.3	158	6.33
RT-DETR	61.4	28.6	79.2	54.5	83.8	50.3	8.33	16.1	91	10.99
SABR-YOLO [49]	87.2	59.2	96.5	62.2	-	-	2.54	-	-	-
LEAD-YOLO [50]	73.5	44.2	92.7	61.5	90.6	58.2	3.15	-	-	-
FFCA-YOLO [51]	73.8	42.7	94.9	62.1	91.0	58.4	2.31	-	-	-
LRFL-YOLO-s [52]	-	-	93.6	64.2	-	-	8.70	20.9	147	-
AFGLFF-YOLO [53]	-	83.1	-	-	-	-	24.8	127.0	-	-
ALF-Net [54]	65.3	33.2	-	-	-	-	3.07	21.8	-	57.9
DMEFF-Net [55]	81.6	52.7	-	-	88.4	-	3.54	-	35	-
EE-YOLO [56]	83.2	58.8	-	-	-	-	9.30			
EFAF-YOLO [57]	77.3	49.6	-	-	-	-	-	-	36	6.00
ESGN-YOLO [58]	76.0	49.2	96.1	62.5	-	-	3.70	12.3	-	-
EF-CoT	**88.6±0.4** (<0.001)	**64.5±0.5** (<0.001)	**97.8±0.3** (<0.001)	**72.5±0.4** (<0.001)	**96.5±0.3** (<0.001)	**67.2±0.4** (<0.001)	3.28	7.5	228	4.39

It is worth noting that the results of both the baseline and EF-CoT are averaged over three independent runs, with standard deviations reported. Paired t-test results show ($p < 0.001$). Performance gains are statistically reliable.

These results reflect the effect of the EF-CoT paradigm. Detection is reformulated as structured modeling and CoT reasoning. Feature representations encode structure, detail, and boundary information. Stepwise reasoning produces stable predictions. Consistent gains across datasets confirm the advantage of reasoning-based detection.

3) Performance Across Different Object Sizes: To further evaluate detection stability under different object scales and different IoU thresholds, this article reports the performance trend from AP50 to AP90 on the RSOD dataset. Compared with evaluation under a single threshold, this experiment describes how model performance changes from loose matching to strict localization.

As shown in Table IV and Fig. 9, the detection performance of both baseline and EF-CoT decreases as the IoU threshold increases from AP50 to AP90. This trend is expected. A

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TABLE III

PERFORMANCE COMPARISON ON THE DIOR DATASET. UNDERLINED VALUES INDICATE DATA DIRECTLY QUOTED FROM THE ORIGINAL SOURCE. THE BEST RESULTS ARE MARKED IN BOLD

Method	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	mAP50
YOLOv5n	82.7	57.3	72.4	61.7	70.4	85.6	35.5	55.7	62.2	96.0	76.3	80.6	87.2	79.3	74.8	78.9	51.5	75.7	57.9	62.3	70.2
YOLOv7-Tiny	80.3	67.1	74.3	63.4	72.6	88.7	45.0	57.3	65.1	92.9	75.7	82.4	91.2	83.4	78.7	82.7	46.5	79.9	55.7	61.1	72.2
YOLOv8n	82.3	67.0	73.9	63.8	74.5	89.1	45.5	60.9	67.4	94.7	78.5	82.6	92.1	84.9	79.6	81.3	54.1	74.4	60.5	66.9	73.7
YOLOv10n	79.2	66.7	69.8	62.0	64.5	87.7	39.7	59.5	62.1	89.8	76.7	84.9	89.2	81.7	79.6	78.3	49.6	77.3	57.3	58.4	70.7
YOLOv11n	80.6	64.4	73.6	64.8	73.3	88.0	41.5	58.2	65.9	91.3	77.8	83.3	89.8	82.6	78.9	81.7	52.5	78.2	58.0	64.8	72.5
YOLOv12n	79.2	64.1	73.3	62.8	73.8	87.1	39.4	58.0	65.6	90.4	75.5	82.0	89.1	82.8	80.0	81.4	50.0	78.2	57.3	60.0	71.5
YOLOv13n	75.3	59.5	70.4	62.8	74.1	84.3	39.7	55.2	66.1	89.5	77.2	77.8	86.1	77.6	78.6	79.7	48.6	74.0	55.6	59.9	69.6
YOLO26n	76.9	63.6	76.7	62.5	70.4	83.9	37.0	57.5	65.8	89.5	76.7	81.7	88.3	77.7	76.2	80.1	49.1	75.4	54.6	62.4	70.3
YOLO-master	81.5	62.8	72.5	65.6	73.8	90.6	36.5	59.1	64.1	90.8	78.7	84.6	91.1	82.1	81.3	80.3	51.4	79.0	56.1	68.1	72.5
Hyper-YOLO-T	84.8	70.9	75.7	66.2	75.8	91.6	42.5	61.1	68.0	**96.0**	78.6	**87.5**	90.7	82.8	79.6	81.5	54.9	82.4	60.5	68.9	75.0
Mamba-YOLO-T	85.9	66.9	76.0	66.2	72.1	91.5	40.0	61.5	67.9	95.6	79.7	83.7	92.5	83.5	80.5	83.8	54.2	79.8	58.3	66.4	74.3
RT-DETR	66.5	72.6	66.4	72.2	82.5	46.2	60.3	63.2	92.7	74.2	75.1	85.6	80.7	78.5	76.6	55.1	76.0	66.5	54.1	63.8	70.8
HAE-Net [59]	82.9	72.3	78.3	67.4	71.7	91.3	49.3	65.0	66.1	92.7	82.3	86.2	90.7	84.9	82.7	83.1	59.4	83.5	64.0	69.2	76.2
RSADet [60]	73.6	**86.0**	72.6	89.6	43.6	75.3	62.3	79.5	68.7	78.6	79.1	57.9	59.2	90.0	55.8	77.0	87.8	65.3	55.3	86.5	72.2
AFDet [60]	82.4	81.5	81.9	89.8	51.7	74.9	58.7	84.2	73.3	79.5	81.0	44.2	62.0	77.8	63.2	76.9	91.0	62.5	59.3	87.1	73.2
CANet [60]	70.3	82.4	72.0	87.8	55.7	79.9	**67.7**	83.5	**77.2**	77.3	83.6	56.0	63.6	81.0	79.8	70.8	88.2	67.6	51.2	89.6	74.3
CSFF [60]	57.2	79.6	70.1	87.4	46.1	76.6	62.7	82.6	73.2	78.2	81.6	50.7	59.5	73.3	63.4	58.5	85.9	61.9	42.9	86.9	68.0
FSoD-Net [61]	88.9	66.9	86.8	**90.2**	45.5	79.6	48.2	**86.9**	75.5	67.0	77.3	53.6	59.7	78.3	69.9	75.0	91.4	52.3	52.0	**90.6**	71.8
DIAG-TR [61]	68.4	72.2	71.9	85.1	29.8	76.8	60.2	70.3	73.2	75.1	35.5	35.5	50.8	41.6	71.8	32.9	81.0	53.1	33.8	78.6	72.0
SESA-Net [61]	93.5	78.2	**91.9**	82.2	46.8	89.3	61.4	67.2	67.5	73.3	76.7	55.8	62.0	78.8	84.8	82.4	89.8	54.4	**76.7**	79.1	74.6
MSNet [61]	**95.3**	67.4	91.0	**90.2**	44.6	82.6	49.2	78.4	64.8	73.4	85.1	66.4	61.8	**94.4**	**90.4**	84.5	**94.8**	48.1	58.9	85.4	75.3
EF-CoT	85.5	69.8	77.3	71.5	**79.6**	**93.0**	51.1	66.1	71.8	95.4	84.5	87.0	**93.7**	88.4	84.1	**86.0**	61.4	**86.9**	64.5	70.4	**78.4**

TABLE IV

DETECTION PERFORMANCE ACROSS VARIOUS OBJECT SIZES ON RSOD

Criteria	Baseline	EF-CoT	AP_{50}	AP_{60}	AP_{70}	AP_{80}	AP_{90}
AP_s	√	-	0.740	0.725	0.487	0.150	0.003
AP_s	-	√	**0.976**	**0.934**	**0.798**	**0.362**	**0.025**
AP_m	√	-	0.953	0.947	0.885	0.657	0.080
AP_m		√	**0.975**	**0.975**	**0.956**	**0.830**	**0.225**
AP_l	√	-	0.954	0.906	0.854	0.678	0.180
AP_l	-	√	0.952	**0.925**	**0.868**	**0.728**	**0.468**

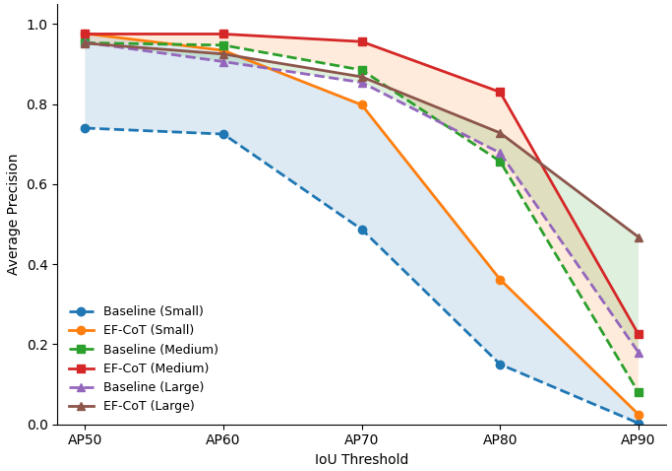

Fig. 9. Performance comparison across different object scales on the RSOD dataset.

lower IoU threshold mainly reflects whether the model can detect the target. A higher IoU threshold puts more emphasis on the overlap between the predicted box and the ground-truth box. Therefore, AP80 and AP90 better reflect boundary localization, scale estimation, and geometric structure modeling. Compared with baseline, EF-CoT shows a more stable performance curve across different IoU thresholds. This result indicates that the proposed method improves not only target discovery but also detection reliability under strict localization criteria.

For small objects, baseline still obtains reasonable performance at low IoU thresholds. However, its performance drops rapidly as the IoU threshold increases. APs decrease from 0.740 at AP50 to 0.003 at AP90. This result shows that baseline can roughly locate small-object regions, but it struggles to regress accurate boundaries. Small objects usually occupy only a few pixels in remote sensing images. Their edge cues are weak, and their texture details are limited. Standard feature fusion can easily bury boundary responses in background noise. In contrast, EF-CoT shows clear gains from AP50 to AP70. It also maintains higher performance at AP80 and AP90. This result shows that the joint modeling of high-frequency detail tokens and edge tokens improves fine-grained structure representation and boundary constraints for small objects.

For medium objects, EF-CoT maintains stable gains at all IoU thresholds. The gains are larger at AP80 and AP90. Medium objects usually contain more complete appearance

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TABLE V
DETECTION PERFORMANCE ACROSS VARIOUS CONFIDENCE THRESHOLDS ON RSOD

Method/Confidence	>=0.1	>=0.2	>=0.3	>=0.4	>=0.5
Baseline	0.599	0.597	0.590	0.588	0.573
EF-CoT	**0.683**	**0.683**	**0.682**	**0.686**	**0.690**

structures. Yet, shadows, roads, building textures, and adjacent objects can still interfere with detection in complex remote sensing scenes. Baseline performance drops at high thresholds. This result shows that its predicted boxes can cover the main object regions, but errors remain in object boundaries, aspect ratios, and center positions. EF-CoT models the overall contour through low-frequency structure tokens. It preserves local details through high-frequency tokens. It constrains geometric boundaries through edge tokens. These cues help predicted boxes keep higher overlap under strict IoU thresholds.

For large objects, baseline and EF-CoT show a relatively small gap at low IoU thresholds. This result indicates that both methods can detect large-scale objects more easily. However, EF-CoT shows a much larger advantage at high IoU thresholds, such as AP80 and AP90. AP90 increases from 0.180 for baseline to 0.468 for EF-CoT. This result shows that the advantage of EF-CoT is not limited to small-object detection. It also improves boundary refinement and spatial structure consistency for large objects. For large objects, the main difficulty is usually not visibility. The key issue is whether the predicted box can accurately cover the complete object boundary. The edge-frequency chain reasoning mechanism in EF-CoT progressively integrates global structure, local detail, and boundary constraints. This process reduces box shifts, boundary truncation, and scale estimation errors.

Fig. 9 further shows that EF-CoT gains more over baseline in high-IoU regions. This pattern indicates that the performance gain does not only come from stronger classification confidence or higher candidate recall. The main gain lies in more accurate spatial localization and boundary regression. If a model only improves target response intensity, the gain usually appears mainly at low-threshold metrics, such as AP50. EF-CoT still keeps clear advantages at AP80 and AP90. This result shows that EF-CoT improves geometric alignment between predicted boxes and ground-truth objects.

Overall, Fig. 9 further verifies the effectiveness of EF-CoT across different object scales and localization strictness levels. Gains at low IoU thresholds show stronger target perception. Stable gains at high IoU thresholds show clear improvements in boundary modeling and precise localization. These improvements mainly come from the reformulation of remote sensing object detection from direct feature mapping to structured token reasoning. Low-frequency tokens provide global structural constraints. High-frequency tokens enhance local detail representation. Edge tokens strengthen boundary localization cues. The step-by-step reasoning mechanism fuses these cues in a complementary manner. Therefore, EF-CoT shows stronger robustness and stability under small objects, complex backgrounds, and high-precision localization requirements.

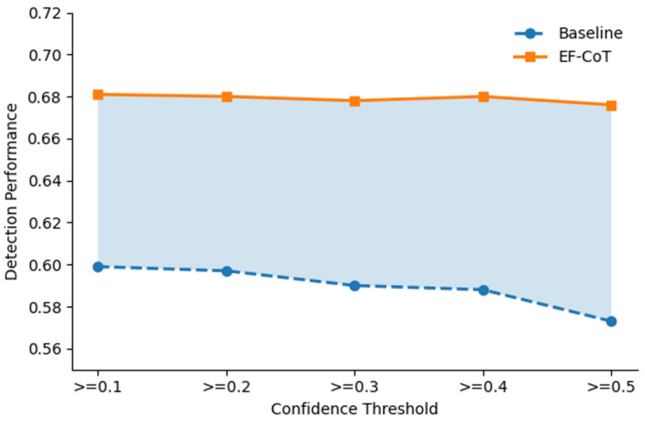

Fig. 10. Detection performance under different confidence thresholds on the RSOD dataset.

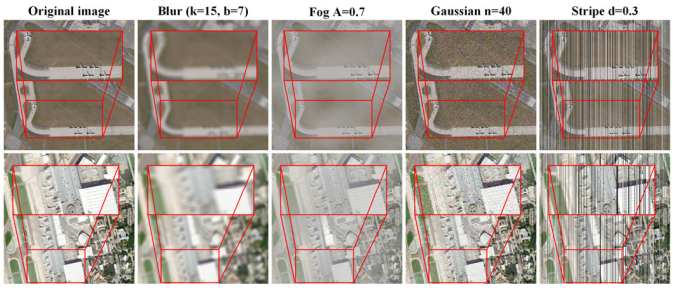

Fig. 11. Different types of degradation images in RS-STOD.

4) Performance Under Varying Confidence Thresholds: Performance is evaluated on RSOD under confidence thresholds from 0.1 to 0.5. Fig. 10 and Table V report the results.

The baseline shows a monotonic decrease as the threshold increases. mAP drops from 0.599 to 0.573. This trend indicates sensitivity to threshold settings. High thresholds remove many medium-confidence predictions. Recall decreases. Overall performance degrades.

EF-CoT maintains stable performance across the full range. mAP remains nearly constant. A slight increase appears at higher thresholds. This behavior indicates a better confidence distribution. Results at all thresholds show EF-CoT outperforming the baseline. The performance curve remains nearly flat. This pattern reflects improved confidence calibration. High-confidence predictions align with true targets. Low-quality predictions are suppressed. At higher thresholds, false positives are reduced. Effective detections are preserved. Precision and recall remain balanced.

Fig. 10 and Table V show improved robustness to threshold variation. Predictions remain consistent under different settings. This stability supports deployment without frequent threshold tuning.

5) Domain-Shift Robustness Evaluation: Robustness is evaluated under a distribution shift on RS-STOD. Fig. 11 shows four degradations: blur, fog, stripe interference, and Gaussian noise. Results are compared with the YOLOv11nbaseline. Table VI shows a consistent trend.

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TABLE VI
DOMAIN-SHIFT ROBUSTNESS EXPERIMENTS FOR OUR METHOD AND YOLOV11N ON RSOD

Blurring (k, b)	Fog (A)	Stripe (d)	Gaussian (n)	mAP_{50} (Baseline)	$\Delta\downarrow$	mAP_{50} (Ours)	$\Delta\downarrow$
-	-	-	-	93.9	0.0	97.8	0.0
(3, 1)	-	-	-	90.8	-3.1	94.9	-2.9
(7, 3)	-	-	-	86.2	-7.7	91.5	-6.3
(15, 7)	-	-	-	79.4	-14.5	87.6	-10.2
-	0.1	-	-	91.7	-2.2	95.7	-2.1
-	0.4	-	-	87.3	-6.6	92.8	-4.9
-	0.7	-	-	81.0	-12.9	88.9	-8.9
-	-	0.1	-	92.5	-1.4	96.2	-1.6
-	-	0.2	-	89.6	-4.3	94.1	-3.7
-	-	0.3	-	85.1	-8.8	90.8	-7.0
-	-	-	10	91.2	-2.7	95.3	-2.5
-	-	-	20	88.4	-5.5	93.6	-4.2
-	-	-	40	83.7	-10.2	89.8	-8.0

TABLE VII
RESULTS COMPARISON OF THE BASELINE COMBINATION WITH DIFFERENT STRATEGIES ON RS-STOD

Method	TGFAENet	GASPPF	C1	C2	C3	C4	C5	mAP_{50}
Baseline	✗	✗	72.1	22.1	83.0	97.6	92.7	73.5
	✓	✗	76.8	25.9	85.6	98.1	94.0	78.2
	✗	✓	74.3	23.4	84.1	97.9	93.2	76.0
	✓	✓	82.6	31.7	89.8	98.7	96.1	88.6

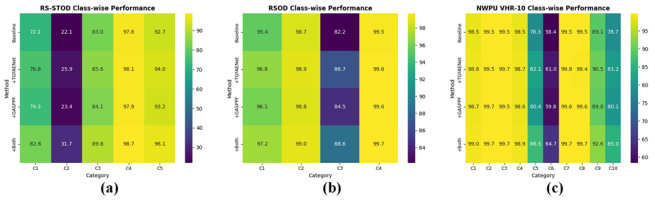

Fig. 12. Heatmap visualization of different categories across datasets. (a) RS-STOD, (b) RSOD, and (c) NWPU VHR-10. The results illustrate the model's response distribution under various remote sensing scenarios, demonstrating its ability to focus on target regions and maintain strong discriminative capability across categories in complex backgrounds.

Performance decreases as degradation increases. EF-CoT maintains higher accuracy under all conditions. The decay rate is slower.

Under blur, kernel size and intensity increase. Baseline drops from 93.9 to 79.4. EF-CoT remains at 87.6. The drop is smaller. This result shows better recovery when structural information is degraded. Under fog, intensity increases from 0.1 to 0.7. Baseline decreases by 12.9. EF-CoT decreases by 8.9. This gap indicates stronger global contrast modeling and resistance to low-frequency interference. Under stripe interference, intensity increases. At d = 0.3, the baseline drops by 8.8. EF-CoT drops by 7.0. The result shows better suppression of structured noise. Under Gaussian noise, both methods degrade. At noise level 20, EF-CoT drops by 4.2. Baseline drops by 5.5. EF-CoT shows higher robustness to random perturbations.

Across all degradations, EF-CoT maintains stable performance. Structured token modeling separates structure, detail, and boundary information. CoT reasoning refines features step by step. Predictions remain consistent under distribution shift. These results indicate stronger robustness in real-world conditions.

E. Ablation Analysis

Ablation studies are conducted on RS-STOD, RSOD, and NWPU VHR-10 to quantify the contribution of each module. TGFAENet and GASPPF are evaluated separately and jointly. Results are reported in Tables VII–IX and Fig. 12.

Overall results show consistent gains when either module is introduced. The largest improvement appears when both modules are combined. This pattern indicates complementary effects.

On the RS-STOD dataset, the baseline obtains 73.5 mAP50. TGFAENet and GASPPF improve mAP50 to 78.2 and 76.0, respectively. Their combination further increases mAP50 to 88.6, with an overall gain of 15.1 points. This result shows that the two modules do not bring a simple linear accumulation. They show clear synergy. TGFAENet focuses on edge-frequency structured token modeling. It strengthens small-target details and boundary representations. However, complex background noise can still disturb high-frequency responses. GASPPF provides a spatial prior. It enhances potential target regions and suppresses irrelevant background responses. After the two modules are combined, GASPPF improves the spatial reliability of token construction. TGFAENet then performs edge-frequency decomposition and step-by-step reasoning. This joint design produces more concentrated target responses, clearer boundary cues, and more stable detection decisions.

On RSOD, the baseline is 93.9. TGFAENet raises it to 95.7. GASPPF raises it to 94.6. The combined model reaches 97.8. TGFAENet contributes larger gains. Improvements are more

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TABLE VIII
RESULTS COMPARISON OF THE BASELINE COMBINATION WITH DIFFERENT STRATEGIES ON RSOD

Method	TGFAENet	GASPPF	C1	C2	C3	C4	mAP_{50}
Baseline	✗	✗	95.4	98.7	82.2	99.5	93.9
	✓	✗	96.8	98.9	86.7	99.6	95.7
	✗	✓	96.1	98.8	84.5	99.6	94.6
	✓	✓	97.2	99.0	88.6	99.7	97.8

TABLE IX
RESULTS COMPARISON OF THE BASELINE COMBINATION WITH DIFFERENT STRATEGIES ON NWPU VHR-10

Method	TGFAENet	GASPPF	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	mAP_{50}
Baseline	✗	✗	98.5	99.5	99.5	98.5	78.3	58.4	99.5	99.5	89.1	78.7	90.4
	✓	✗	98.8	99.5	99.7	98.7	82.1	61.0	99.8	99.4	90.5	81.2	92.1
	✗	✓	98.7	99.7	99.5	98.6	80.4	59.8	99.6	99.6	89.8	80.1	91.0
	✓	✓	99.0	99.7	99.7	98.9	86.5	64.7	99.7	99.7	92.6	85.0	96.5

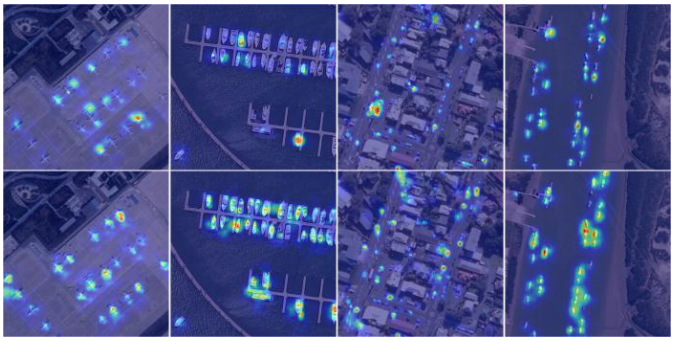

Fig. 13. Heatmap visualization comparison in the CoT reasoning ablation study (from top to bottom) without CoT and with EF-CoT. The results show that introducing EF-CoT leads to more focused attention on object regions, reduced background interference, and better alignment with object structures.

visible in complex categories. GASPPF stabilizes predictions. The joint model achieves the best result. On NWPU VHR-10, the baseline is 90.4. TGFAENet increases it to 92.1. GASPPF increases it to 91.0. The combined model reaches 96.5. Larger gains appear in categories with complex structures. This pattern indicates stronger discriminative ability under diverse conditions.

Category-level analysis explains these trends. TGFAENet improves fine-grained structure modeling through edge-frequency decomposition and structured tokens. Small-object representation becomes stronger. GASPPF introduces Gaussian spatial priors and multiscale context. Target regions receive higher responses. Background interference is reduced.

The combined model performs both structure modeling and spatial guidance. Robustness improves across scales and complex backgrounds. Fig. 12 provides visual confirmation.

These results show that TGFAENet and GASPPF improve performance from two aspects: structure modeling and spatial priors. Their combination produces consistent and large gains in complex remote sensing scenes.

1) Ablation on CoT Reasoning: To verify the effectiveness of the chain reasoning mechanism in the overall framework, this article further conducts a capacity-controlled comparison on the RSOD dataset. This experiment also excludes the possibility that the performance gain only comes from increased model capacity. Table X reports the quantitative results. Fig. 13 shows the visualization results. The experiment includes three variants: Direct Concat, Capacity-matched Direct Fusion, and EF-CoT. Direct Concat directly concatenates the generated low-frequency features, high-frequency features, and edge texture features. It does not introduce explicit reasoning states. Capacity-matched Direct Fusion adds extra convolution layers on top of Direct Concat. Its parameter count and computational cost are close to those of EF-CoT. However, it still does not contain intermediate reasoning states, such as evidence extraction, relation modeling, and decision fusion. EF-CoT adopts the proposed explicit chain reasoning mechanism.

The quantitative results show a clear trend. Direct Concat has 2.74M parameters and 6.6 GFLOPs. It obtains 93.9% mAP50 and 62.4% mAP50-95. After capacity-matched direct fusion is introduced, Capacity-matched Direct Fusion increases the parameter count and computational cost to 3.26M and 7.4 GFLOPs. These values are close to the 3.28M parameters and 7.5 GFLOPs of EF-CoT. Its mAP50 and mAP50-95 increase to 94.8% and 65.7%, respectively. This result shows that larger model capacity can improve the direct fusion baseline to some extent. However, under nearly the same parameter count and computational cost, EF-CoT further achieves 95.7% mAP50 and 68.9% mAP50-95. It outperforms Capacity-matched Direct Fusion by 0.9 and 3.2 points, respectively. These results show that the advantage of EF-CoT does not simply come from more parameters or higher computation. The main gain comes from the structured modeling ability of the explicit chain reasoning path.

From the perspective of model design, Direct Concat and Capacity-matched Direct Fusion still belong to the direct feature fusion paradigm. Capacity-matched Direct Fusion increases model capacity through extra convolution layers. Yet, it still mixes low-frequency structure, high-frequency detail, and edge texture information at an early stage. It lacks explicit intermediate reasoning states and step-by-step relation

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TABLE X
CAPACITY-CONTROLLED COMPARISON BETWEEN DIRECT FUSION AND EF-COT REASONING ON RSOD

Method	Reasoning State	Params↓	GFLOPs↓	mAP_{50}↑	$mAP_{50\text{-}95}$↑
Direct Concat	✗	2.74	6.6	93.9	62.4
Capacity-matched Direct Fusion	✗	3.26	7.4	94.8	65.7
EF-CoT	✓	3.28	7.5	95.7	68.9

TABLE XI
ABLATION STUDY ON TGFAENET ON RSOD

Method	mAP_{50}	$mAP_{50\text{-}95}$	AP_s	AP_m	AP_l
High Frequency Branch	94.3	64.8	0.862	0.951	0.924
Low Frequency Branch	93.8	63.5	0.835	0.958	0.930
Edge Extraction Branch	94.6	65.9	0.881	0.953	0.928
TGFAENet	**95.7**	**68.9**	**0.901**	**0.964**	**0.936**

modeling. This direct fusion process can couple structural information, texture details, and background noise. It can then limit localization accuracy in complex remote sensing scenes. In contrast, EF-CoT explicitly models the structured input state, evidence extraction state, and structural relation state. It progressively refines token representations through evidence extraction, relation modeling, and decision fusion. This process allows the model to enhance target evidence and model cross-token relations before final prediction.

The heatmaps in Fig. 13 further support this conclusion. Compared with direct fusion models without explicit reasoning, EF-CoT generates more concentrated and structurally consistent response distributions. It reduces invalid activations in background regions. It also aligns feature responses better with target regions and boundary structures. This shift from diffuse responses to structured focus shows that EF-CoT does not rely only on larger model capacity. The explicit reasoning process organizes features more effectively and suppresses noise more reliably.

In summary, the capacity-controlled ablation experiment shows that EF-CoT still outperforms direct fusion methods under similar parameter counts and computational costs. This result verifies the effectiveness of the chain reasoning mechanism over direct recognition or direct fusion. Its main advantage lies in explicit intermediate states and a step-by-step reasoning path. These designs improve the structural quality of feature representations and the reliability of the decision process. They also strengthen detection ability in complex remote sensing scenes.

2) Ablation on Triple-Branch Structured Tokens: To validate the effectiveness of the proposed three-branch structured token modeling strategy, a systematic ablation study is conducted on the RSOD dataset. The high-frequency branch, the low-frequency branch, the edge extraction branch, and the full TGFAENet are evaluated. Results are reported in Table XI.

From an overall perspective, a single branch is insufficient to handle multiscale object representation.

The high-frequency branch shows strong performance in detail modeling. It reaches 94.3 in mAP50. However, mAP50-95 and small-object accuracy remain limited. This result indicates that high-frequency signals capture texture details but lack global structural constraints. The low-frequency branch focuses on semantic-level representation. It performs better on large objects. Performance drops on small objects. This behavior shows weak fine-grained structural modeling. The edge branch explicitly encodes boundary information. Among single-branch settings, it shows the most balanced performance. This result highlights the importance of boundary cues in remote sensing structures. However, edge-only modeling cannot integrate frequency-domain semantics, which limits its capacity.

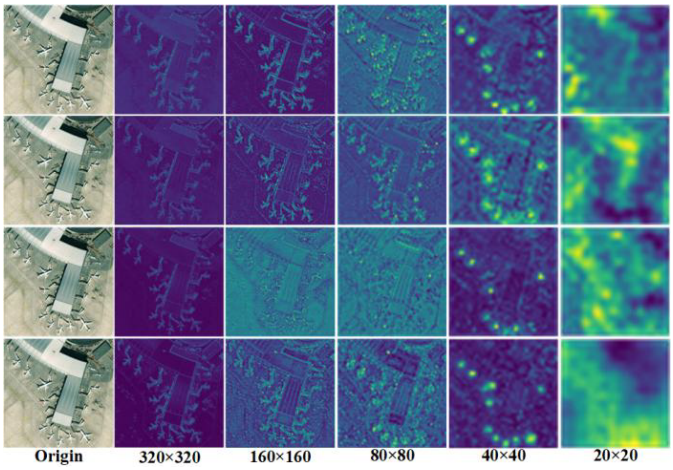

Fig. 14. Visualization comparison of multiscale feature maps for four variants in the TGFAENet ablation study (from top to bottom) high-frequency branch, low-frequency branch, edge extraction branch, and TGFAENet. The results demonstrate that integrating multifrequency information with edge modeling produces more complete feature responses and clearer structural representations, significantly improving small-object and fine-detail perception.

After combining the three branches in TGFAENet, performance increases significantly. The model reaches 95.7 in mAP50 and 68.9 in mAP50-95. Small-object detection also improves over the best single branch. This result shows that frequency information and edge modeling complement each other. Token representations become more expressive under joint modeling, which improves multiscale perception.

Fig. 14 provides visual evidence. High-frequency features contain rich details but also show noise in local regions. Low-frequency features show smooth regions but weak boundaries. Edge features strengthen object contours but lack semantic context. TGFAENet produces more complete activation patterns. Responses align better with small objects and fine structures.

To analyze the effect of different edge extraction operators in the edge branch, experiments are conducted on RSOD. Canny,

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TABLE XII
COMPARISON OF DIFFERENT EDGE EXTRACTION METHODS ON RSOD

Method	mAP_{50}	$mAP_{50\text{-}95}$	AP_s	AP_m	AP_l
Canny	94.2	64.7	0.865	0.950	0.926
Laplacian	93.9	63.8	0.842	0.949	0.925
Sobel	**94.6**	**65.9**	**0.881**	**0.953**	**0.928**

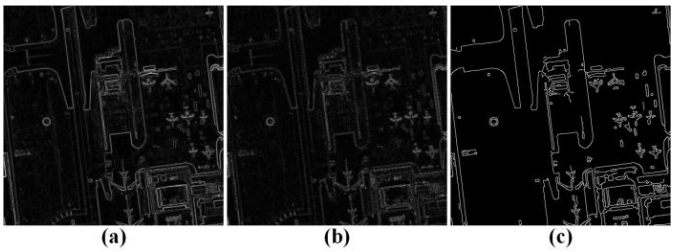

Fig. 15. Comparison of three classical edge detection methods. (a) Sobel, (b) Canny, and (c) Laplacian. The methods exhibit different characteristics in edge localization accuracy, noise suppression, and detail preservation.

Laplacian, and Sobel edge detectors are compared. Results are reported in Table XII.

Sobel edge detector achieves the best overall performance. Small and medium objects show the greatest improvement. This result indicates that first-order gradient information provides more stable structural cues in remote sensing scenes. Canny edge detector shows slightly lower performance. Edge localization is accurate in many cases. However, weak structural regions are partially missed. This leads to incomplete structural representation. The Laplacian edge detector shows the lowest performance. The method is more sensitive to noise. False responses increase in complex backgrounds. This reduces the reliability of edge features in structured token construction.

Fig. 15 shows the visualization results of different edge operators. Sobel produces more continuous boundaries. Noise responses are also more controlled. This pattern shows a better balance between edge continuity and noise suppression.

Canny shows discontinuous edge responses in several regions. Some object boundaries appear fragmented. Laplacian produces stronger responses in noisy backgrounds. Nonobject regions also contain noticeable activations.

Overall, Sobel provides more stable edge representations. This behavior makes it more suitable as the basic operator for the edge branch. The three-branch structured Token representation becomes more stable, and detection performance improves accordingly.

Further visualization analysis is conducted on RS-STOD for objects at different scales, as shown in Fig. 16.

The baseline model shows clear scale sensitivity in complex scenes. Small-object regions exhibit high miss rates. Dense regions also contain more false detections. These results indicate that conventional feature modeling lacks stable structural representation under scale variation and background interference.

In contrast, the proposed method shows more robust and consistent detection results under the same scenes. Both small dense objects and medium-to-large objects are more completely covered. False positives and missed detections are both reduced.

Fig. 16. Detection results for objects at different scales on the RS-STOD dataset. (Top row) Baseline model and (bottom row) proposed method. Green boxes indicate correct detections. Red boxes indicate missed detections. Blue boxes indicate false detections. The proposed method reduces both missed detections and false detections.

This improvement is linked to the three-branch structured token design. High-frequency information supports fine-detail perception. Low-frequency information provides semantic consistency. Edge information strengthens structural boundaries. These components jointly improve feature discrimination across scales.

Fig. 16 confirms the effectiveness of the proposed design in complex remote sensing scenes. Recall for small objects is improved. Background false alarms are reduced. These results further support the superiority of the method in multiscale object detection tasks.

3) Ablation on GASPPF: To evaluate the effectiveness of each component in the proposed GASPPF spatial prior module, three ablation settings are designed on the RSOD dataset. These include: 1) GA only; 2) GASPPF without GA; and 3) the full GASPPF. Results are reported in Table XIII.

Overall results show that using GA alone yields the lowest performance. Local Gaussian weighting strengthens local responses, but no structural prior is introduced. Feature representations remain unstable under complex backgrounds.

After removing GA, performance increases noticeably. Spatial prior modeling improves feature selectivity. Target regions receive more focused responses. Background interference is reduced. This setting shows that spatial prior structure contributes more to stable feature learning than isolated attention weighting.

The full GASPPF achieves the best performance. Small-object detection improves the most. The combination of spatial priors and GA enhances local structure representation. This joint effect improves fine-grained perception.

Fig. 17 provides visual evidence. GA-only features show scattered activation regions. The "without GA" setting produces more stable distributions but weaker fine details. Full GASPPF produces compact responses with clearer boundaries. Small objects and fine structures show stronger activations.

Overall, GASPPF improves feature consistency and discriminability through the joint modeling of spatial priors and attention mechanisms.

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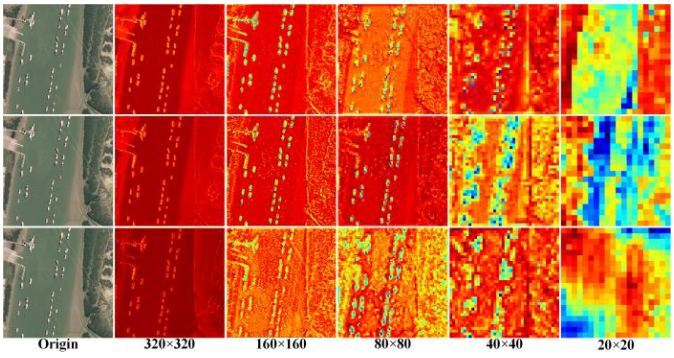

Fig. 17. Visualization comparison of feature maps at different scales for three variants in the GASPPF spatial prior ablation study: (top) GA, (medium) GASPPF without GA, and (bottom) GASPPF. The results show that incorporating spatial priors and GA progressively enhances feature representation, especially for small objects and fine-grained structures.

TABLE XIII
ABLATION STUDY ON THE GASPPF SPATIAL PRIOR MODULE ON RSOD

Method	mAP_{50}	$mAP_{50\text{-}95}$	AP_s	AP_m	AP_l
Gaussian Attention (GA)	93.7	63.6	0.848	0.950	0.928
GASPPF Without GA	94.1	64.8	0.862	0.952	**0.930**
GASPPF	**94.6**	**65.9**	**0.881**	**0.953**	0.928

TABLE XIV
CROSS-ARCHITECTURE GENERALIZATION RESULTS ON RS-STOD

Methods	Venue	Backbone	EF-CoT	$mAP_{50}\uparrow$	$mAP_{50\text{-}95}\uparrow$
FasterNet [62]	CVPR'23	CNN	✗	74.6	46.8
FasterNet	CVPR'23	CNN	✓	**76.2**	**50.1**
LSKNet [63]	ICCV'23	CNN	✗	75.3	47.5
LSKNet	ICCV'23	CNN	✓	**77.1**	**51.2**
DEIMv2 [64]	arXiv'25	Transformer	✗	71.6	45.3
DEIMv2	arXiv'25	Transformer	✓	**73.8**	**48.6**
RT-DETR [11]	CVPR'24	Transformer	✗	61.4	28.6
RT-DETR	CVPR'24	Transformer	✓	**64.2**	**32.1**
VMamba [65]	ICML'24	Mamba	✗	76.3	49.4
VMamba	ICML'24	Mamba	✓	**78.0**	**52.6**
Mamba-YOLO [66]	AAAI'25	Mamba	✗	74.8	48.1
Mamba-YOLO	AAAI'25	Mamba	✓	**76.6**	**51.5**

4) Cross-Architecture Generalization: To evaluate whether the proposed EF-CoT paradigm generalizes across architectures, experiments are conducted on multiple mainstream detection frameworks, including CNN, Transformer, and Mamba-based models. The CNN group includes FasterNet and LSKNet. The Transformer group includes DEIMv2 and RT-DETR. The Mamba group includes VMamba and Mamba-YOLO. All models are trained under identical settings to ensure fair comparison.

Table XIV shows consistent performance gains across all architectures. In CNN-based models, FasterNet increases from 74.6 to 76.2 in mAP50, and LSKNet increases from 75.3 to 77.1. The mAP50-95 gains are +3.3 and +3.7, respectively. These results show improved structural representation in local feature modeling.

In Transformer-based models, DEIMv2 and RT-DETR follow the same trend. RT-DETR increases from 28.6 to 32.1 in mAP50-95. This result indicates that explicit reasoning improves structural consistency in global modeling frameworks.

In Mamba-based models, VMamba shows +1.7 in mAP50 and +3.2 in mAP50-95. Mamba-YOLO shows +1.8 and +3.4, respectively. These gains confirm effectiveness under state-space sequence modeling.

Across all architectures, EF-CoT consistently improves detection performance. This pattern indicates that structured token reasoning is compatible with different backbone designs.

EF-CoT shows consistent performance gains across different architectures. Improvements are more pronounced on mAP50-95. This pattern indicates better localization accuracy and improved boundary alignment.

This behavior is linked to the multistage reasoning process in token space. Feature representations are refined step by step. Coarse responses gradually converge into structurally consistent representations. This process improves compatibility across different network backbones.

Overall, cross-architecture results show that EF-CoT is not tied to a specific detection framework. It functions as a general structured reasoning mechanism. Stable improvements are observed in CNN, Transformer, and Mamba-based models. This confirms the effectiveness of shifting from feature modeling to reasoning-based modeling.

Fig. 18. Typical failure cases of the proposed method on the NWPU VHR-10, RSOD, and RS-STOD test sets. Red boxes indicate missed detections. Purple boxes indicate false detections. Orange boxes indicate duplicated bounding boxes.

F. Failure Case Analysis

Although the proposed method achieves stable and strong performance across multiple remote sensing datasets, a finer-grained error analysis still reveals limitations under extremely complex scenes. Fig. 18 presents qualitative failure cases on NWPU VHR-10, RSOD, and RS-STOD. Red boxes indicate missed detections. Purple boxes indicate false positives. Orange boxes indicate duplicate detections.

Overall results show that missed detections are mainly concentrated in dense small-object scenes and low-contrast conditions. In RS-STOD and RSOD, some extremely small objects with background similarity are not activated by the model. This behavior indicates weak feature responses under low signal-to-noise conditions, even with the three-branch structured token design.

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False positives (purple boxes) mainly appear in regions with complex backgrounds or strong texture interference. This issue is more evident in RS-STOD. Background textures are sometimes activated as object-like responses. This result reflects incomplete alignment between frequency features and spatial features across branches. Duplicate detections (orange boxes) occur mainly when object boundaries are unclear or objects are closely spaced. This pattern is more frequent in NWPU VHR-10. It indicates limited suppression of redundant proposals before NMS, especially in crowded regions.

Overall, these failure cases show that the proposed method improves multiscale detection through structured tokens and spatial priors. However, extremely small-object conditions, strong background interference, and dense object layouts still remain challenging. Future improvements can focus on stronger cross-scale consistency constraints and finer structural suppression mechanisms to further improve robustness and boundary discrimination.

V. Conclusion

This article addresses missed detections of small targets, false positives in complex backgrounds, and unstable boundary localization in remote sensing object detection. It proposes an edge-frequency chain reasoning framework, termed EF-CoT. Unlike traditional detectors that directly map features to detection results, EF-CoT decomposes detection into three stages: structured token construction, edge-frequency relation modeling, and step-by-step decision generation. It explicitly builds a reasoning path in the feature space. This path consists of evidence extraction, relation modeling, and decision fusion. This design allows the model to decompose and reorganize low-frequency structure, high-frequency detail, and edge constraints before detection. It improves feature representation and decision stability in complex remote sensing scenes.

For model implementation, this article designs TGFAENet as a network instance of EF-CoT. TGFAENet constructs structured tokens with explicit physical semantics through a low-frequency structure branch, a high-frequency detail branch, and an edge constraint branch. It also uses an edge-guided interaction mechanism to strengthen the association between high-frequency responses and object boundaries. The role of this module is not to simply increase feature fusion channels. It converts structural information, detailed information, and geometric boundary information into intermediate representations for reasoning. This design improves small-target recognition and boundary localization.

This article also builds GASPPF to enhance spatial prior modeling. GASPPF aggregates multiscale context through a spatial pyramid structure. It uses Gaussian spatial attention to continuously weight potential target regions. This process suppresses irrelevant responses in complex backgrounds. TGFAENet focuses on edge-frequency token reasoning. GASPPF focuses on spatial prior constraints. The two modules are functionally complementary. TGFAENet improves structured evidence representation. GASPPF improves the reliability of spatial responses. Ablation results show that their joint use produces stronger synergy. This result shows that structured token reasoning and spatial prior modeling work well together for complex remote sensing object detection.

Experiments on RS-STOD, RSOD, NWPU VHR-10, and DIOR verify the effectiveness of the proposed method. Compared with several mainstream detectors, EF-CoT shows more stable detection performance in mAP, mAP50-95, different object scales, and high-IoU thresholds. The gains are more evident in dense small-target scenes and complex-background scenes. Further ablation experiments show that TGFAENet, GASPPF, and the edge-frequency chain reasoning mechanism all contribute to the final performance gains. Visualization results also show that the proposed method generates more concentrated target responses, clearer boundary activations, and fewer background interference responses.

This article still has several limitations. First, the current framework mainly targets horizontal bounding-box detection in remote sensing images. Its adaptation to rotated objects and arbitrarily oriented objects still requires further validation. Second, EF-CoT currently models RGB remote sensing images. Multispectral, SAR, infrared, and other multisource remote sensing information have not yet been fully used. Third, the model achieves a good balance between accuracy and efficiency. Yet, its deployment performance on edge devices and large-scale real-time remote sensing applications can still be further optimized. Future work will focus on rotated bounding-box detection, multimodal remote sensing reasoning, and lightweight deployment. These directions will further improve the generalization and practical value of EF-CoT in real remote sensing applications.

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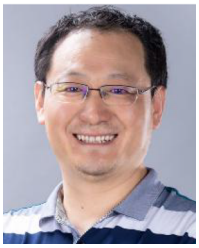

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